Basic income versus fairness: redistribution with inactive agents

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Abstract

I study redistribution between inactive, unemployed, and employed agents who are heterogeneous in their preferences and their productive skills in the labor market and at home. The government considers that redistribution across skills is desirable, but redistribution across preferences is not: if everyone has the same preferences, equality is optimal, but if everyone has the same skills, *laissez-faire* is optimal. In turn, this leads to a simple sufficient statistic to assess any reform: any extra dollar of transfer should go to the agent whose consumption is the furthest below what she would have consumed if everyone had average skills. In high-income countries, the gap between the lowest consumption and the average productivity is empirically much larger in the labor market than among inactives. Hence, introducing an inactivity benefit, or basic income, would neither be welfare-improving nor optimal for that social welfare function, regardless of labor supply elasticities.

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1 Introduction

By law, voluntarily inactive agents are typically ineligible for social assistance programs because these transfers are conditioned on labor market participation in developed countries.[1] However, in recent years, this standard scheme has been criticized by those advocating for the introduction of a universal basic income, which is a social benefit granted on an individual basis without a means test nor any work requirements.[2] In particular, introducing a basic income would amount to granting some positive transfers to able-bodied inactives, i.e. an inactivity benefit.[3]

As with any transfers, a basic income may bring equity gains by reducing the welfare inequality between actives and inactives, but it comes with some efficiency costs as it disincentivizes job-seeking efforts of unemployed agents as well as work effort provided by employed individuals. While the equity gains of a basic income have not received much attention, its efficiency costs are subject to considerable disagreement in the literature: labor supply elasticities and the associated efficiency costs of transfers are found to be large in Conesa et al. (2023), Daruich and Fernández (2024), and Golosov et al. (2024) but small in Cesarini et al. (2017). In this paper, I measure the equity gains of introducing a basic income *whatever the size of its efficiency costs*.

In the model, agents decide whether to be active or inactive, and how many hours to work when active. Most importantly, they differ in their productivity in labor market and at home as well as in their preferences. This allows to capture relevant features of the policy question, because a government transfer will have redistributive consequences not only across different skills as in the standard Mirrlees (1971) model, but also across agents with heterogeneous disutility of participation and preference for leisure. Yet, these multiple dimensions of heterogeneity bring a serious difficulty: how can we compare the welfare of agents with heterogeneous tastes and aggregate them in a social welfare function? In other words, how can we measure equity gains?

I build social welfare based on three key axioms. First, I impose the standard Weak Pareto principle. Second, the government does not wish to redistribute across heterogeneous preferences: if everyone has the same skills, the optimal policy is *laissez-faire*, i.e. the absence of redistribution, even if there remain consumption inequalities due to unequal preferences. Third, the government wishes to redistribute across skills: redistributive transfers between agents with identical preferences but unequal skills improve

[1] This is the case for all social assistance programs for the 29 developed economies studied in this paper. An overview of the legislation can be found in the MISSOC (2021) database. We come back on enforcement issues later in the paper.

[2] Academic supporters include notably Atkinson (1996) or Van Parijs and Vanderborght (2017) and the policy movement includes the Basic Income Earth Network (BIEN) (https://basicincome.org/faqs/).

[3] Because of this equivalence, the paper uses the terms 'basic income' and 'inactivity benefit' interchangeably.

social welfare. This compensation (for one's skills)-responsibility (for one's preferences) approach, or *equality of opportunity*, has been pioneered by Fleurbaey and Maniquet (2006, 2007, 2011, 2018) for the standard income tax problem.

Because the government considers that redistribution across skills is desirable but redistribution across preferences is not, it will ask a counterfactual question: how much would each agent consume if inequalities in skills were eliminated and everyone had the *average* productivity in the labor market and at home? The difference between that benchmark and her actual consumption, or money-metric utility, measures how much of her current situation reflects bad luck in skills rather than a choice due to preferences. Because of inequality aversion in skills, the government wishes to reduce this difference, which leads to a social welfare function that *maximins* a money-metric utility.

When applied to the redistribution problem between actives and inactives, this social welfare function reduces to a sharp and simple criterion. The government will compute the gap between the worst consumption at home and the average home production, and compare it to the gap between the worst consumption in the labor market and the average wage. The main theorem of the paper shows that any welfare-improving reform consists in reducing the largest of these two gaps. As a consequence, the government should increase the inactivity benefit (resp. social assistance) whenever the difference between the worst consumption and the average production among inactives (resp. actives) is larger than among actives (resp. inactives).

This implies that the desirability of an inactivity benefit does not depend on the consumption inequalities *between* actives and inactives, but rather *among* actives and *among* inactives. Intuitively, this comes from the fact that, once inequalities *among* actives and *among* inactives vanish, all residual consumption inequalities *between* them are entirely due to unequal preferences and, therefore, are unproblematic for the government.

Because the theorem holds regardless of the government's budget constraint and for any labor supply elasticities, it is only a statement about the desirability of a reform, not its fiscal feasibility. Hence, this theorem amounts to a sufficient statistic to assess the equity gains of any tax-benefit reform. Conveniently, this sufficient statistic holds whether the current tax system is optimal or not[4], whether the tax reforms under consideration are local or global, and whatever the empirical correlations between skills and preferences.

Once we know how to find welfare-improving reforms, we can also study optimal

[4]Second-best optimum might be unreachable for real-world tax-benefit systems because of political economy constraints (Bierbrauer et al., 2021), or more generally because actual governments do not behave like the Mirrleesian planner (Stantcheva, 2016). Hence, policy recommendations of practical use are more likely to emerge from the study of welfare-improving reforms.

policies. In particular, I show that optimal transfers should be designed such that the gap between the worst consumption and the average production among is equal to the gap among inactives.[5] This uniquely pins down the relative level of transfers between actives and inactives, that is the level optimal inactivity benefit for a given level of optimal social assistance.[6] In turn, if optimal social assistance increases by a dollar and the gap among actives shrinks by a dollar, then the optimal inactivity benefit should also increase by a dollar to restore the equal-gap condition. This suggests that basic income should supplement rather than crowd out existing transfer programs.

Would the introduction of a basic income be desirable? It is ultimately an empirical question. In the last part of the paper, I study whether a dollar of inactivity benefit would be welfare-improving or optimal in 29 high-income countries. To compute the sufficient statistics, one only needs the average wage, the level of social assistance (or other safety net programs), and the gap between the lowest and average home production. I elicit conservative bounds on home production by exploiting data from the *Global Survey on Working Arrangements* (G-SWA) on time savings when working from home (Aksoy et al., 2023). I combine these estimates with data on current tax-benefit systems for childless singles and lone parents (OECD, 2020).

Even under a series of conservative assumptions, the empirical application finds that the worst inactive is closer to the average home production than the worst active is to the average wage. This holds whether I consider the worst consumption when active to be a full-time minimum wage worker or a social assistance recipient. As a result, all 29 governments should first increase transfers to actives before any dollar spent on an inactivity benefit constitutes a welfare improvement. This is equivalent to saying that introducing a basic income would not be welfare-improving in all those countries.

Next, I quantify lower bounds on these increases in transfers to actives to justify any dollar of basic income. I find that these magnitudes are almost always unrealistically large: on average, governments should at least triple the level of social assistance before any dollar of basic income is welfare-improving. In sum, the inquiry shows that either the overall amount of social transfers is much too low in developed economies, or granting an inactivity benefit cannot be welfare-improving.

Overall, this paper suggests that the equity gains of granting some benefits to vol-

[5]By contradiction, if the gap, say, in the labor market was larger than at home at the optimum, then a cut of one dollar in inactivity benefit combined with an increase of a dollar in social assistance would be welfare-improving, which contradicts optimality.

[6]Their joint absolute value can only be obtained by taking a stance on the efficiency costs of transfers and adding a government budget constraint.

untarily inactive agents are tenuous. This demonstrates a normative tension[7] between basic income[8] and equality of opportunity. This holds against a series of conservative assumptions. In particular, the government does not have a preference for labor market production over home production. This also holds independently of the efficiency costs of transfers, that is whether labor supply elasticities are large or small and for any revenue requirement.

Literature

Assessing the desirability of basic income with respect to equality of opportunity is particularly relevant for several reasons[9]. First, recent surveys have suggested that equality of opportunity has received some public support (Saez & Stantcheva, 2016; Stantcheva, 2021; Weinzierl, 2017). Second, these axioms are rooted in a long tradition in political philosophy (Dworkin, 1981; Fleurbaey, 2008; Rawls, 1971; Van Parijs, 1995, 2021). Third and most importantly, philosophers sharing closely related ethical standpoints disagree on the policy recommendations that it entails[10]. Rawls (1988) argued that his difference principle does not justify that *Malibu surfers should be fed* as they enjoy so much leisure that they are not among the worst-off. Van Parijs (1991) countered that one should not take a stance on what a good life is, and granting a basic income would allow anyone to enjoy as much leisure as one wishes, including the worst-off, thereby providing Rawlsian justification to basic income.

The study of conditionality of welfare benefits has a long history in economics[11] and basic income has been recently studied by e.g. Banerjee et al. (2019), Conesa et al. (2023), Daruich and Fernández (2024), Ghatak and Maniquet (2019), Golosov et al. (2024), and Hoynes and Rothstein (2019). However, most papers consider utilitarian welfare or do not exploit heterogeneous tastes. The main contribution of the present paper is to derive a simple and estimable sufficient statistic to guide policy-making even

[7]However, this tension is not an impossibility. In particular, holding agents responsible for their preferences may be normatively questionable in light of the literature on behavioral welfare economics (Bernheim, 2009; Bernheim & Taubinsky, 2018). Appendix C considers that some agents suffer from a stigma cost of participating in the labor market (as in Besley and Coate (1992a) and Moffitt (1983)) that the government compensates for. I show that if governments are ready to pay on average three times their current safety net in order to compensate for the welfare recipient stigma, introducing a basic income may be a welfare-improving reform.

[8]I only study the desirability of the conditionality of social benefits to labor market participation, while basic income proposals additionally require waiving conditionalities to means and to the household composition (Van Parijs, 1995). Arguably, the desirability of an inactivity benefit is a first step for the study of the desirability of a fully fledged basic income.

[9]Let me note that I do not aim to endorse a single view of social welfare but rather to link a practical policy recommendation with transparent ethical underpinnings.

[10]Notoriously, neither Rawls nor Dworkin endorsed themselves the view that an inactivity benefit would be justified by their theories of justice. As Dworkin (2000) puts it "*forced transfers from the ant to the grasshopper are inherently unfair*" (p.329).

[11]See Besley and Coate (1992b, 1995), Boadway and Cuff (2014), Boadway et al. (2003), and Boone and Bovenberg (2013) among many others.

under multidimensional heterogeneity in preferences, home production and wages. This is used to characterize the difference of transfers that should hold between unemployed and inactives.

The paper also contributes to optimal taxation theory. In the canonical Mirrlees (1971) model, agents may only react to tax-benefit reforms by decreasing their hours worked i.e. on the intensive margin. This class of models has been amended to allow for participation decisions in Choné and Laroque (2005, 2011), Diamond (1980), Hansen (2021), Jacquet et al. (2013), and Saez (2001, 2002). However in these *pure* extensive margins models, there is no difference between an unemployed and an inactive. Several papers have then added search frictions to rationalize involuntary unemployment together with endogenous participation decisions (Hungerbühler & Lehmann, 2009; Hungerbühler et al., 2006; Jacquet et al., 2014; Lehmann et al., 2011). Yet, they do not allow the government to distinguish the transfers it gives to the inactive from the one to the unemployed.

Boadway and Cuff (2018) allow the government to differentiate the transfers to nonparticipants from the transfers to the (involuntary) unemployed. Nonetheless, their model assumes homogeneous preferences, no intensive margin and a piece-wise linear income tax schedule while the present paper relaxes all these three assumptions. Kroft et al. (2020) also operates this differentiation and do not have any of the aforementioned shortcomings. Let me pinpoint three main differences with that paper. First, they model wages as endogenously determined in general equilibrium while in the present paper, wages are left exogenous. Second, they have a (Bergson-Samuelson) weighted utilitarian social welfare function whereas I consider an (Arrovian) social ordering function that reflects the ethics of equality of opportunity. Third, they focus on the derivation of the optimal tax system but I will also look at welfare-improving reforms, even in a suboptimal world. As I model a home sector[12] and a formal sector, the paper is related to the Mirrleesian optimal tax derivation of Rothschild and Scheuer (2013) in the multi-sector Roy (1951) model. The important difference is that they assumed that the tax schedule is uniform across sectors while I do not. In particular, it will be assumed that outcomes from the home sector are unobservable, and that the government can only give a lump-sum amount to all inactives. Beaudry et al. (2009) also assume that home production is unobservable but hours worked are observed and they focus on the difference between social assistance and unemployment insurance.[13] In contrast here, I focus on the difference between social assistance and inactivity and assume that gross income

[12]Gayle and Shephard (2019) introduced home production in an optimal income tax model. However, their focus is different as they estimate a large structural microeconometric model, with a marriage market and they focused on the jointness of spouses taxation.

[13]For a more recent treatment of the redistribution versus insurance problem in cash transfers and unemployment insurance, see Ferey (2022).

are observed along with the activity binary decision.

The observability of the binary activity decision supposes that there exists a screening mechanism that the government uses to differentiate inactives from actives. In the main body, this screening mechanism is costless and perfect but we discuss the relevance of the findings for the real world where that device is costly and fuzzy in Sections 5.1 and 6. The question of the design of ordeals is left aside and has been treated by e.g. Kleven and Kopczuk (2011) and Rafkin et al. (2024).

I outline two additional differences with standard optimal tax papers. First, the literature since Saez (2001, 2002) typically studies small local tax reforms. By contrast, this paper derives a sufficient statistic that can assess any tax reform, including (potentially suboptimal) large and global ones such as the introduction of a basic income. Second, this paper does not need to impose any structural assumption on preferences[14] nor any correlation between heterogeneity dimensions, while optimal tax papers typically do so to solve the multidimensional screening problem. Both facts are possible thanks to the axiomatic derivation of a social objective as a transitive ordering between any two allocations i.e. any two tax-benefit systems. This endeavor has been inspired by the fair income tax literature (Fleurbaey & Maniquet, 2006, 2007, 2011, 2018). The present paper also contributes to the latter by including inactive agents as well as additional dimensions of heterogeneity and deriving the axiomatic characterization in this new environment.

In section 2, I formalize the environment. In section 3, I construct the social objective in the first-best. In section 4, I introduce the second-best environment where the government cannot observe skills and preferences but can tag transfers to income and labor market participation. I derive the main theoretical results. In section 5, I present the empirical application. In section 6, I conclude.

2 Model

There is a finite set $\mathcal{I} = \{1, 2, ..., I\}$ of agents whose generic element is denoted by i. There are only two goods, consumption and labor, forming a bundle denoted by $(c_i, l_i) \in X$. The homogeneous consumption good $c_i \in \mathbb{R}_+$ is produced either in the home sector or in the labor market. The labor supply variable l_i is set to -1 when the agent stays at home, or takes value of hours worked in a normalized interval $[0, 1]$ when the agent is in the labor market. Hence, an inactive agent has $l_i = -1$, an unemployed agent has $l_i = 0$ and an employed agent has $l_i \in (0, 1]$.

[14] Dworczak et al. (2023) mechanism design exercise finds a sufficient condition for the transfers to those meeting ordeals to be larger than transfers to those not meeting ordeals if preferences are quasilinear. In the present paper, there are no quasilinear restrictions. Moreover, the optimal transfers exercise precisely quantifies the optimal difference of transfers between inactives and unemployed.

Each agent is endowed with a convex preference ordering $\succsim_i$ that can be represented by a continuous ordinal utility function $u_i(c_i, l_i)$ which is strictly increasing in c_i and non-increasing in l_i. An agent may have an idiosyncratic disutility of participation[15] to the labor market whenever $u_i(c_i, -1) - u_i(c_i, 0) \geq 0$ which captures the utility loss for an inactive that becomes unemployed while keeping the same level of consumption. Importantly, the disutility of participation must be distinguished from the willingness to work.[16] In this framework, disutility of participation embodies a preference to produce at home rather than in the labor market, while willingness to work reflects the substitutability between consumption and hours worked on the labor market.

In addition to their preferences, agents are also heterogeneous along their vector of innate productive abilities $(w_i, h_i) \in [\underline{w}, \bar{w}] \times [\underline{h}; \bar{h}] \subseteq \mathbb{R}^2_+$, where the first coordinate denotes the marginal productivity used on the labor market and the second coordinate captures the *surplus* of home production that inactivity allows for relative to activity, not its *level*.

In the labor market, I retain the standard assumption of a constant return to scale technology whose sole input is hours worked, and its marginal productivity is given by w_i. In the home sector, h_i captures the surpluses of production that active agents loose by joining the labor market. It is a reduced-form term for outcomes of activities such as gardening, child rearing or housekeeping. Obviously in reality all agents produce at home, be they inactive, unemployed or employed. Hence, it is implicitly assumed here that all actives produce an identical level at home, normalized to 0 (i.e. the first-best Laissez-faire consumption of unemployed agents) and inactives produce h_i more than them[17].

In the first-best, the second fundamental welfare theorem prescribes that efficient redistribution could be achieved through lump-sum transfers. Denoting these transfers by t_i, the first-best budgets, illustrated in Figure 1, are defined by :

$$B(t_i, w_i, h_i) = \Big\{ (c_i, l_i) \in X : c_i \leq a(l_i) w_i l_i + (1 - a(l_i)) h_i + t_i \Big\}$$

where $a(\cdot)$ is an indicator function that takes value 1 if the agent is active (i.e. $l_i \in [0, 1]$)

[15]It is known that the disutility of participation displays substantial heterogeneity in the cross-section of households (Kaplan & Schulhofer-Wohl, 2018). Here, it may capture (but it is not restricted to) the stigma utility cost of welfare conditionality as in Friedrichsen et al. (2018) and Moffitt (1983), see Appendix C. .

[16]In this setup, the willingness to work may be approximated by the marginal rate of substitution over non-negative values for l. A low (resp. high) marginal rate of substitution in absolute value reflects a high (resp. low) willingness to work

[17]In the empirical section 5, I argue that, while absolute levels of home production are typically unobservable, reasonable bounds for home surpluses h_i may be found.

or 0 if the agent stays at home (i.e. $l_i = -1$)[18].

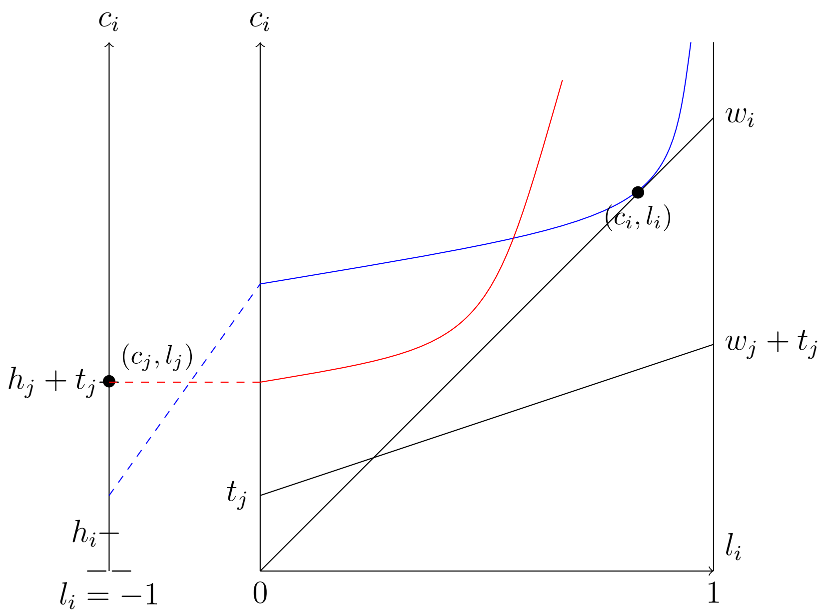

Figure 1: Illustration of the model for two agents i, j. Blue agent i receives no lump-sum transfer and has a positive disutility of participation, but still chooses optimally to be active. Contrarily, the red agent j receives $t_j > 0$ and does not display disutility of participation but chooses to be inactive.

Observe that when $a(\cdot) = 1$, this model nests the standard linear production model used e.g. in optimal taxation theory. Hence, agents can react to tax reforms along three margins in this model: (i) intensive (by decreasing hours worked when employed), (ii) standard-extensive (by switching from employed to unemployment), and (iii) participation-extensive (by switching from activity to inactivity). Four important remarks must be raised.

First, this model allows for both involuntary and voluntary unemployment. In the former case, the (primitive) state of the labor market nullifies the productivity of some agents (such that $\underline{w} = 0$) which can remain active with $l_i = 0$. By contrast, voluntary unemployment arises when $l_i = 0$ is the utility-maximizing choice of an agent endowed with some positive wage rate $w > 0$[19].

Second, I assume that there is no intensive margin in the home sector[20]. Of course,

[18]Observe that the use of this indicator function renders moot the actual value of l_i when inactive as long as it is not in [0,1]. The choice of -1 is both arbitrary and harmless.

[19]In the first-best, unemployment (both voluntary and involuntary) only happens in the knife-hedge case where $h = 0$. In the second-best, unemployment may happen when transfers to unemployed are larger than transfers to inactives, which is the empirically relevant case.

[20]Moreover, an intensive margin in the home sector would imply that the inactivity benefit distorts effort in the home production which would be unfavorable to the emergence of a basic income. Hence, this modelling assumption is conservative with respect to my result.

in reality agents partition their time between leisure, paid work and home production. However, labor market inactivity is a binary status for any tax-benefit system, such that there is no such thing as a part-time inactive in the eyes of the fiscal authority: agents can either be full time in the home sector or not at all in this sector.

Third, inactive and unemployed agents both enjoy the full unit of leisure. However, only the former are able to produce at home. This is equivalent to saying that there is a fixed time cost spent looking for a job when unemployed that can be used productively when inactive. Hence, h_i must be understood (and measured) as the product of this time cost with the hourly idiosyncratic[21] value of production in the home sector. For example, in a legal working week of, say, 40 hours, the inactive and the unemployed do not provide any hours worked and thus enjoy 40 hours of leisure. However, an unemployed must spend, say, 10 hours sending job applications during which the inactive takes care of his children. The inactive's h_i is then the product of $\frac{10}{40}$ with the shadow price of an hour of day care services in that economy. This estimation procedure for h_i is discussed at length in section 5 below.

Fourth, the model assumes away any positive externality of job search. This is again a conservative assumption, in the sense that such externality would lay the grounds for Pigouvian subsidies to the unemployed (or Pigouvian tax on inactivity) in the redistributive problem, which would be unfavorable to the emergence of a basic income. Despite that, I find below that basic income is not desirable such that this result would be even stronger with such an externality.

Note that throughout the paper, no parametric specification of utility functions is imposed. Moreover, I will not impose any correlation between primitives $\succsim_i$, w_i and h_i, neither at the individual level nor in the cross-section. As a consequence, the results are valid *for any* empirical correlations observed in the data.

Finally, let me define an economy e by a list of endowments for each agent in each heterogeneity dimension $e = \{(\succsim_i, w_i, h_i)\}_{i \in \mathcal{I}}\}$. I denote the set of all such economies by E. An allocation is a bundle for each individual and is denoted $(c, l) = \{(c_i, l_i)_{i \in \mathcal{I}}\} \in X^{\mathcal{I}}$.

3 Fair social objective

3.1 Characterization

Each economy can yield many allocations which are associated with observable inequalities in consumption-labor outcomes (c_i, l_i) all originating from unobserved heterogeneity in primitives $(\succsim_i, w_i, h_i)$. The key question for a government is: when should an alloca-

[21] In a structural macro exercise with time use data, Boerma and Karabarbounis (2021) show that (1) inequalities in the home sector are quantitatively important and (2) inequalities in home production efficiency are needed to explain the variance of home inputs conditional on wages and preferences.

tion (c, l) be socially preferred to an allocation (c', l')?

There are infinitely many ways to answer this question. The standard approach used in economics in general and in optimal taxation theory in particular consists in using a weighted sum of utility functions to assess welfare improvements. However, when agents have heterogeneous preferences, such summation is sensitive to the cardinalization of preferences, i.e. to the choice of the particular $u_i(\cdot)$ function that represents $\succsim_i$.

There is no commonly admitted way to choose such a cardinalization. Yet, it implicitly embodies a way to compare welfare interpersonally, and as such it is ethically-loaded. The literature in normative economics has acknowledged that policy recommendations are contingent on an ethical view and as such it is better to visibilize it rather than leaving it to the researcher's choice. But how can we link ethics with a social welfare function?

A popular method in recent years has been to rely on Saez and Stantcheva (2016)'s generalized weighted sum of utilities where weights can be a function of $(\succsim_i, w_i, h_i)$ to reflect different fairness views. While this approach is flexible and tractable, it has a major drawback: it may fail transitivity in its policy recommendation (Sher, 2024).

In the present paper, I build an Arrovian social welfare function, i.e. a function that associates to each economy a transitive ordering of allocations.[22] In particular, among the many possible social welfare functions, I will select the one that respects three key fairness principles: (1) **Weak Pareto**, (2) **Responsibility** and (3) **Weak Transfer**.

Weak Pareto ensures that a Pareto-dominated allocation will never be chosen by the planner. Yet, there are typically many points on the Pareto frontier, and the next two axioms single out the one that reflects equality of opportunity. **Responsibility** requires that in the knife-edge case where all agents have the very same productive endowments w and h, then the *Laissez-faire* allocation (i.e. the absence of transfers) is the social optimum, even if unequal $\succsim_i$ still generates inequalities. As a result, **Responsibility** does not fight inequalities spawned by unequal preferences. Conversely, **Weak Transfer** requires that if two agents have identical preferences and identical labor supply decisions (i.e. identical l_i), a transfer from the richer to the poorer agents is a social improvement. Hence, **Weak transfer** fights inequalities spawned by unequal skills. As argued above, this ethical view may be relevant for the policy debate on basic income.

I now present the social welfare function that pursues the ethics of equality of opportunity: it maximins $M_i(c_i, l_i)$. In short, $M_i(c_i, l_i)$ is a money-metric utility function that measures the well-being of agent i when she consumes the bundle (c_i, l_i) as the transfer

[22]This aggregation from individual preferences to social welfare follows the Arrow (1950) tradition and constructs the social welfare function axiomatically as in the pioneering work of Fleurbaey and Maniquet (2007).

that renders this agent indifferent between (c_i, l_i) and her preferred bundle in the budget determined by the average productive vector. Loosely speaking, the further below the individual sees herself from an average agent, the worst will be her well-being.

Definition 1. *The equality of opportunity social welfare function considers that (c, l) is socially strictly preferred to (c', l') whenever*

$$\min_{i \in \mathcal{I}} M_i(c_i, l_i) > \min_{i \in \mathcal{I}} M_i(c_i', l_i')$$

where $M_i(c_i, l_i)$ is a money-metric utility function defined as

$$M_i(c_i, l_i) = \min t \qquad \textit{such that } (c_i, l_i) \sim_i \max_{\succsim_i} B(t, \tilde{w}, \tilde{h})$$
$$\textit{where} \quad \tilde{w} = \frac{1}{I} \sum_i w_i \qquad \tilde{h} = \frac{1}{I} \sum_i h_i \qquad \textit{and } t \in \mathbb{R}$$

The graphical construction of $M_i(c_i, l_i)$ is illustrated in Figure 2 for the case of a two-agent economy that was introduced in Figure 1.

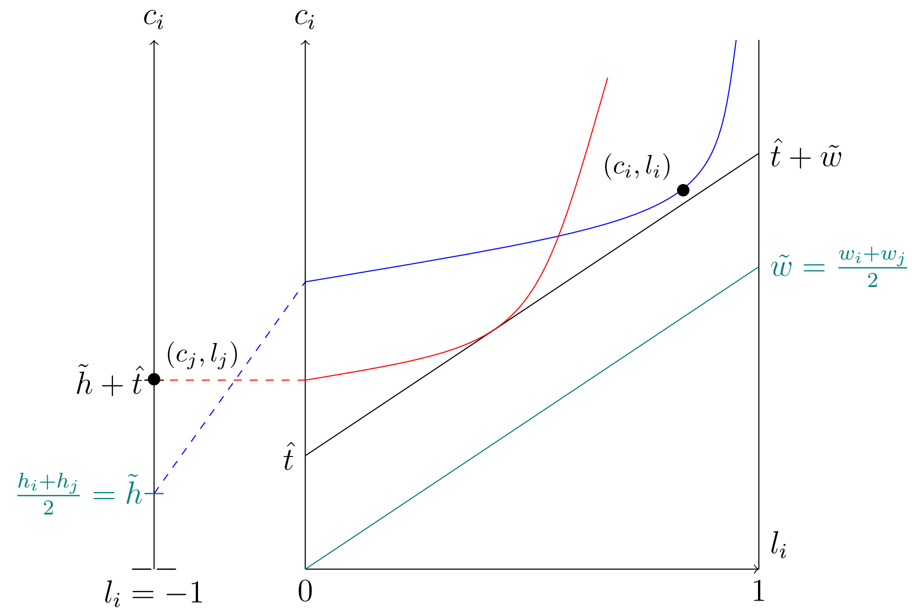

Figure 2: Construction of the money-metric welfare $M_i(c_i, l_i)$ using the example economy of Figure 1. First, one computes $(\tilde{w}, \tilde{h})$ as $(\frac{w_i + w_j}{2}, \frac{h_i + h_j}{2})$. This yields the green reference budget $B(0, \tilde{w}, \tilde{h})$. Then, one finds the welfare of an agent by computing the smallest transfer t that translates the reference budget up to tangency (or intersection) with the indifference curve. In this case, blue agent i's indifference curve at (c_i, l_i) intersects $B(0, \tilde{w}, \tilde{h})$ when $l_i = -1$ so $M_i(c_i, l_i) = 0$, while red agent j's indifference curve at (c_j, l_j) is tangent to $B(\hat{t}, \tilde{w}, \tilde{h})$ such that $M_j(c_j, l_j) = \hat{t} > M_i(c_i, l_i)$ and agent i is the worst-off.

Definition 1 implies that the social welfare function will focus on the agent having the smallest $M_i(c_i, l_i)$, i.e. the agent with the lowest well-being. In other words, the social objective is of the *maximin* type.

The next theorem shows that the aforementioned three axioms single out the social welfare function in Definition 1 when they are combined with two technical axiom. The fourth axiom, **Mean-Preserving Separability** is reminiscent of the Separability axiom used in virtually all social welfare function since Fleming (1952): indifferent agents should not matter in the social decision.[23] The last axiom, **Hansson (1973) Independence**, allows the social welfare function to use indifference curves rather than mere utility levels to escape Arrow (1950)'s impossibility. All formal definitions of axioms are relegated to Appendix A.1.

Theorem 1. *If a social welfare function satisfies* ***Weak Pareto****,* ***Responsibility****,* ***Weak Transfer****,* ***Mean-Preserving Separability*** *and* ***Hansson Independence****, then it is the equality of opportunity social welfare function in Definition 1.*

Proof. See Appendix A.2. ■

Before providing intuition behind this result, it is useful to study its consequences for optimal policy. What would be an optimal schedule of lump-sum transfers in the first-best with respect to this social welfare function? Because of the maximin nature of the social objective, optimality in the first-best is achieved whenever all agents in the economy have an identical well-being measure, i.e. $M_i(c_i, l_i) = M_j(c_j, l_j)$ for all $i, j \in \mathcal{I}$. All agents then reach their indifference curve tangent to this reference budget set[24], thereby all enjoying the same level of welfare in the eyes of the planner. I illustrate a first-best optimal allocation in Figure 3.

Two crucial points must be raised. First, observe that given the heterogeneity in preferences, equality of welfare at the optimum does not mean that all agents consume the same bundle.[25] Second, and most importantly, at the optimal allocation there are no productive inequalities *within* sectors, but only *between* sectors. More precisely, all indifference curves either pass through the inactive agents' consumption or are tangent to the same hourly wage over $l \in [0, 1]$. Yet, the government does not wish to redistribute further to equalize consumption across sectors, even if home production is not equal to hourly wages. This shows that a government that pursues equality of opportunity wishes to redistribute within sectors, not between sectors. Intuitively, this comes from the fact that, once within-sector inequalities vanish, the sectoral decision is entirely driven by

[23]The axiom used in this paper is logically weaker than the standard Separability axiom. This weakening is required to avoid an impossibility, as is proven in the Appendix A.

[24]This tangency condition of the optimal allocation is different from (and compatible with) the tangency condition of individuals' constrained utility maximization. For example, in Figure 3, agent j optimizes by choosing (c_j, l_j) which implies that her indifference curve is tangent to her individual budget when $l_j > 0$, but it coincides with the green reference budget $B(m, \tilde{w}, \tilde{h})$ when $l = -1$.

[25]Also, note that such strongly egalitarian allocations may also be reached with a standard utilitarian setup in the first-best, as has been known since Edgeworth (1897).

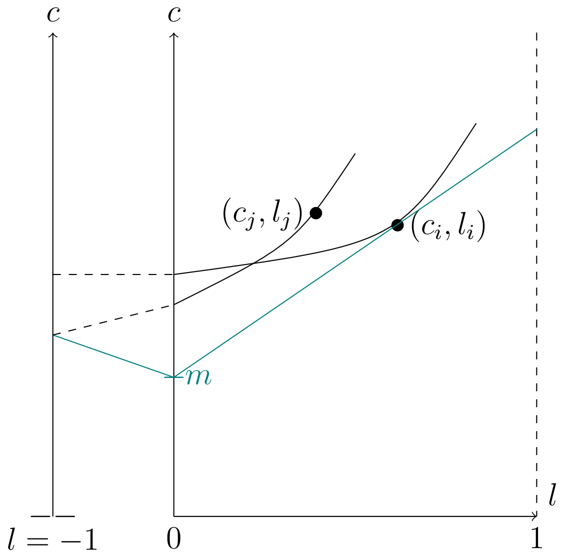

Figure 3: An allocation (c, l) equalizing $M_i(c_i, l_i) = M_j(c_j, l_j) = m$ across agents is optimal with respect to the *equality of opportunity* social welfare function in the first-best. All indifference curves are tangent (or intersect) with the green reference budget set $B(m, \tilde{w}, \tilde{h})$.

preferences and the government does not wish to redistribute across preferences.

3.2 Why the average?

Theorem 1 asserts that the fair social objective should be (1) of the maximin type,(2) over money-metric utilities, and (3) with the particular money-metric utility function $M_i(c_i, l_i)$. If one compares these results to the literature, only the latter contains some novelty. Indeed, it is now standard in the literature on equality of opportunity that the finite inequality aversion embodied in **Weak transfer** axioms turns out to be infinite when combined with the other axioms.[26] Moreover, it comes at no surprise that money-metric utilities are a useful and flexible way to cardinalize utilities when preferences are heterogeneous.[27] What is peculiar and new with Theorem 1 is that the reference prices in the money-metric utility are pinned down by arithmetic averages $\tilde{h}$ and $\tilde{w}$. I now sketch an example to provide intuition for this result. It is illustrated in Figure 4.

Consider four agents: two agents with preferences $\succsim_a$ and two with preferences $\succsim_b$. For the purpose of this example, let us assume that these are purely inelastic preferences, i.e. the labor supply choice is independent of their endowments. We know that the social ordering should be *maximinning* a money-metric utility function and wonder which reference endowments are pinned down by the axioms.

[26] See e.g. Fleurbaey and Maniquet (2011) and Piacquadio (2017) for a discussion.

[27] In related papers, Bosmans et al. (2018), Fleurbaey and Maniquet (2017), and Piacquadio (2017) provided recent axiomatization of money-metrics as arguments of social welfare functions. Their history dates back at least to Samuelson and Swamy (1974) and recent papers showed their empirical usefulness when agents have heterogeneous preferences (Baqaee et al., 2024; Jaravel & Lashkari, 2024)

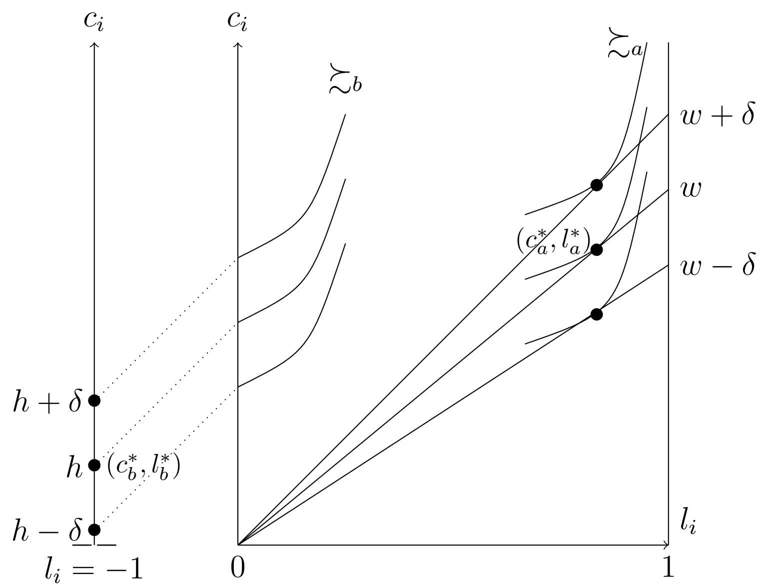

Figure 4: Why the average? By **Responsibility**, (c^*, l^*) is optimal if everyone has the same (w, h) endowment. By **Weak Transfer**, (c^*, l^*) is also optimal for arithmetic mean-preserving spreads from the equal endowment economy.

When these four agents have identical skills, the economy is $\{(\succsim_a, h, w), (\succsim_a, h, w), (\succsim_b, h, w), (\succsim_b, h, w)\}$. In such case, **Responsibility** implies that the absence of redistribution is optimal and *Laissez-faire* allocation (c^*, l^*) is optimal. Because of the maximin nature of the social objective, we must have that the money-metric welfare of all agents is equal at the optimal allocation. Which reference endowments in the economy $\{(\succsim_a, h, w), (\succsim_a, h, w), (\succsim_b, h, w), (\succsim_b, h, w)\}$ ensures that the money-metric welfare of agents a and b is equal at (c^*, l^*)? The answer is: all those that are such that

$$R(\{h, w\}, \{h, w\}, \{h, w\}, \{h, w\}) = \{h, w\}$$

where R is the reference operator. This equation tells us that $R(\cdot)$ is a mean. At this stage though, it could be many means: the quadratic mean, the cubic mean and all other Hölder means respect that equation.

Now imagine that there is a shock to the endowments of agents and the economy becomes $\{(\succsim_a, h+\delta, w+\delta), (\succsim_a, h-\delta, w-\delta), (\succsim_b, h+\delta, w+\delta), (\succsim_b, h-\delta, w-\delta)\}$ where $\delta > 0$. In other words, among each pair of agents with preferences $\succsim_a$ and $\succsim_b$, there is one winning (δ, δ) in her endowments and one loosing (δ, δ). In that case, **Weak Transfer** imposes that a within-pair transfer from the fortunate to the unfortunate is a social improvement. Iterating **Weak Transfer**, one finds that the social optimum in the new economy is also the allocation (c^*, l^*).

We have shown that the social optimum is identical in two economies, where one economy is constructed as an arithmetic mean-preserving spread from the other. This forces the reference price to be the same in both economies i.e.

$$R(\{h+\delta, w+\delta\}, \{h-\delta, w-\delta\}, \{h+\delta, w+\delta\}, \{h-\delta, w-\delta\}) = R(\{h, w\}, \{h, w\}, \{h, w\}, \{h, w\})$$

There is only one mean operator $R(\cdot)$ that satisfies the last two equations: the arithmetic average. This shows why arithmetic averages as reference prices are compatible with the axioms of **Weak Transfer** and **Responsibility**.

The formal proofs relegated in the Appendix A.1 build on this intuition. In Theorem A.1 I prove that there an impossibility between **Weak Transfer**, **Responsibility** and **Separability**. Indeed, removing indifferent agents from the economy is not innocuous in our setting because they change the size of the pie, which influences the potential redistribution that the axioms of equality of opportunity will be able to achieve. In the proof of Theorem 1, I show that **Mean-Preserving Separability** is enough to escape the impossibility, i.e. restricting the Separability axiom to case that do not affect the per-capita amount of resources in the economy.[28]

4 Redistributive taxes and transfers

In this section I characterize the tax-benefit system pursuing equality of opportunity under incentive-compatibility constraints. As in Mirrlees (1971), these constraints arise because the government is unable to observe the endowment vector $(\succsim_i, w_i, h_i)$ of each individual.

Moreover, the government is unable to observe l_i and can only observe $y_i = a(l_i)w_i l_i$, the gross labor income reported in tax returns. I will also assume that, when $y_i = 0$, the government can perfectly distinguish the inactive from the unemployed agent. In other words, $a_i(\cdot)$ is observable even if l_i is not. This is a key assumption: it allows the government to differentiate the transfers it gives to unemployed agents from transfers to inactives. This is consistent with the observation that in most developed economies, there exists a screening mechanism enforcing the conditionality of welfare benefits to a job-seeking behavior.[29]

For active agents with $a(l_i) = 1$, the government designs the tax schedule on the labor market through the nonlinear tax function $\tau(y)$ which is a subsidy whenever $\tau(y) < 0$

[28]Formally, this does not prove that arithmetic averages are the only reference prices compatible with **Weak Transfer** and **Responsibility**. Such a proof is beyond the scope of the present paper. Yet, it can be proven that popular alternatives to the arithmetic mean will violate some axioms, in particular the geometric and harmonic means, the $\min$ and the $\max$ operator. All results in section 4 are valid for a general reference vector $(\tilde{h}, \tilde{w})$. In the empirical section 5, I benefit from the fact that arithmetic averages for wages are readily available statistics in many countries.

[29]In practice, such screening mechanism may be imperfect and costly. I come back in the conclusion on the sensitivity of the results to this assumption.

on some y. For agents out of the labor market with $a(l_i) = 0$, the government cannot observe home production yet these inactive agents may receive a subsidy $D \in \mathbb{R}$ which is a tax if $D < 0$ such that the second-best budgets are

$$B(\tau, D, w_i, h_i) = \left\{ (c_i, l_i) \in X | c_i \leq a(l_i)[y_i - \tau(y_i)] + (1 - a(l_i))[h_i + D] \right\}$$

I call D the inactivity benefit and $-\tau(0)$ the social assistance. Agents choose l_i to maximizes $u_i(c_i, l_i)$ under this second-best budget constraint.

One may wonder about the relationship between the tax-benefit system (τ, D) studied here and the basic income proposals. Indeed, why wouldn't we give a basic income *universally* to both actives and inactives? Observe that the tax-benefit system (τ, D) studied here is completely equivalent to a universal basic income $(\tau' - D, D)$ for a τ' chosen such as $y - \tau'(y) + D = y - \tau(y)$, i.e. both systems decentralize the very same allocation. In other words, it is the (consequentialist) difference between active and inactive transfers that has welfare consequences, not the (deontological) means of transfers[30]. For clarity, I retain the formulation with an inactivity benefit.

An incentive-compatible allocation (c, l) decentralized by the tax-benefit system (τ, D) is such that each agent gets their preferred bundle in their budget $B(\tau, D, w_i, h_i)$:

$$\forall i, j \in \mathcal{I}, (c_i, l_i) \succsim_i (c_j, l_j) \text{ or } (c_j, l_j) \notin B(\tau, D, w_i, h_i)$$

Before turning to the main results, we need one assumption. It ensures that there exists at least one agent in the economy that has the worst endowments and *hardworking* preferences, i.e. no preference for leisure and a disutility of participation such that she is indifferent between social assistance and inactivity.

Assumption (Hardworking Poor Existence)**.** *For any incentive-compatible allocation (c, l) decentralized by (τ, D) and any economy $e \in E$, there exists an agent $j \in \mathcal{I}$ such that $(w_j, h_j) = (\underline{w}, \underline{h})$ and $(\underline{h} + D, -1) \sim_j (-\tau(0), l_j)$ for all $l_j \in [0, 1]$.*

In other words, this Hardworking Poor agent has preferences such that she is indifferent between all the points of the budget $B(\tau, D, \underline{w}, \underline{h})$. This assumption may seem strong for small economies but is arguably more reasonable when designing tax-benefit systems for large economies, as the present paper is concerned about. Moreover, since governments cannot observe indifference curves, it seems prudent to design policies as

[30]In layman terms, whether a job-seeker receives 1000 USD as social assistance or 500 USD as basic income combined with 500 as social assistance yield the same bundle for that jobseeker.

if any type of agents existed, should that indeed be the case in reality.[31]

The next theorem translates the allocation ordering implied by the axioms from a function expressed in abstract well-being measures (in Theorem 1) to a function expressed in terms of policy tools and economy's parameters. It is stated and proven for the case where the smallest wage rate in the population is 0, i.e. $\underline{w} = 0$. It can be interpreted either as agents unable to work or as involuntary unemployed agents.[32] The case $\underline{w} > 0$ can be derived in a similar way and is relegated to Appendix B.1.

Theorem 2. *Under Hardworking Poor Existence, for $e \in E$ such that $\underline{w} = 0$, and any two incentive-compatible allocations (c, l) and (c', l') decentralized by (τ, D) and (τ', D') respectively, one has that (c, l) is socially preferred to (c', l') with respect to the equality of opportunity social welfare function whenever*

$$\min\left\{\underline{h} + D - \tilde{h}; -\tau(0) - \tilde{w}\right\} > \min\left\{\underline{h} + D' - \tilde{h}; -\tau'(0) - \tilde{w}\right\} \tag{1}$$

Proof. Consider the tax-benefit system (τ, D) that decentralizes (c, l). By Hardworking Poor Existence, $\exists j \in \mathcal{I}$ with $(w_j, h_j) = (\underline{w}, \underline{h}) = (0, \underline{h})$ with preferences such that she is indifferent between all feasible bundles for her. At the allocation (c, l), that Hardworking Poor agent receives a bundle $(c_j, l_j) \in B(\tau, D, 0, \underline{h})$ and the her indifference curve through (c_j, l_j) passes through each point of the budget $B(\tau, D, 0, \underline{h})$. We illustrate these preferences in Figure 5.

Using Definition 1, we can compute the money-metric welfare of that agent j. Graphically, it consists in finding the smallest transfer t that translates the green budget $B(0, \tilde{w}, \tilde{h})$ to make it intersect with the red indifference curve of agent j. Algebraically, this is

$$M_j(c_j, l_j) = \min_{\hat{l}} t \text{ s.t. } \begin{cases} -\tau(0) = \tilde{w}\hat{l} + t & \text{if } \hat{l} \in [0, 1], \\ \underline{h} + D = \tilde{h} + t & \text{if } \hat{l} = -1 \end{cases}$$

Because of Hardworking Poor agent j's flat indifference curve when active, the smallest value of t when $\hat{l} \in [0, 1]$ can only be found when $\hat{l} = 1$. Hence, the money-metric utility of j reduces to

$$M_j(c_j, l_j) = \min\left\{\underline{h} + D - \tilde{h}; -\tau(0) - \tilde{w}\right\}$$

[31] Because Hardworking Poor Existence forces a match between preferences and the tax-benefit system under consideration, it restricts the set of tax-benefit systems that can be studied. In particular, it cannot be that $-\tau(0) < D + \underline{h}$ because there is no negative disutility of participation that could render an agent indifferent between social assistance and inactivity in such case. I argue that this is inconsequential because, in such a tax system where no one chooses social assistance $-\tau(0)$, it is completely free for the government to raise social assistance until $-\tau(0) = D + \underline{h}$.

[32] Involuntary unemployment corresponds to the case in which the state of the labor market nullifies an agent's productivity.

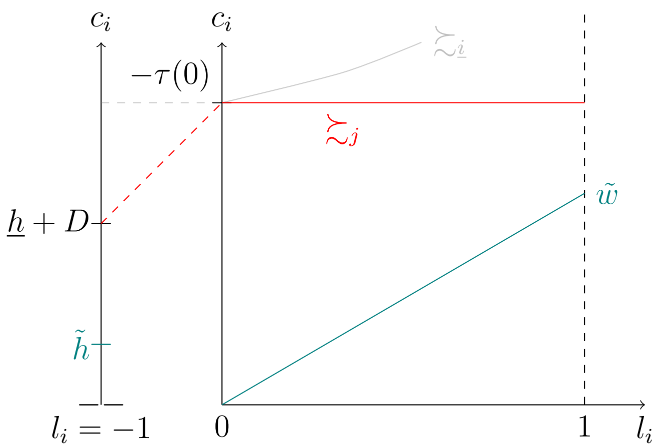

Figure 5: Illustration of the proof of Theorem 2. Hardworking Poor Existence guarantees the existence of the red indifference curve of agent j that is indifferent between social assistance and the worst consumption when inactive. Because $\underline{w} = w_j = 0$, that agent cannot do better than $-\tau(0)$ in the labor market. The welfare of agent j is the transfer that makes her indifferent to the green budget $B(0, \tilde{w}, \tilde{h})$, i.e. the smallest transfer that translates the green budget to intersect with the red indifference curve.

Now, consider a generic agent $\underline{i}$ with the worst endowments $(0, \underline{h})$ who chooses her preferred bundle over $B(\tau, D, 0, \underline{h})$, but that agent $\underline{i}$ has different preferences than the Hardworking Poor j. By revealed preferences, the indifference curve of $\underline{i}$ at that optimal choice must be pointwise above[33] the budget set $B(\tau, D, 0, \underline{h})$, as illustrated by the optimal choice of the gray agent in Figure 5. Importantly, observe that the budget set $B(\tau, D, 0, \underline{h})$ coincide with the red indifference curve. As a consequence, the smallest transfer required to make gray agent $\underline{i}$ indifferent to the green reference budget $B(0, \tilde{w}, \tilde{h})$ needs to be larger than the smallest transfer required for red agent j, so that we have

$$M_j(c_j, l_j) \leq M_{\underline{i}}(c_{\underline{i}}, l_{\underline{i}}) \quad \forall \underline{i} : (w_{\underline{i}}, h_{\underline{i}}) = (0, \underline{h}) \qquad \text{and } (c_j, l_j), (c_{\underline{i}}, l_{\underline{i}}) \in B(\tau, D, 0, \underline{h})$$

This proves that the Hardworking Poor preferences are the worst possible preferences in terms of welfare $M_i(c_i, l_i)$ among the set of agents with the lowest skills.

Now, all agents i with $(w_i, h_i) \geq (0, \underline{h})$ have a budget $B(\tau, D, w_i, h_i)$ that contains $B(\tau, D, 0, \underline{h})$. Hence, by incentive-compatibility, $(c_i, l_i) \succsim_i (c_j, l_j) \in B(\tau, D, 0, \underline{h})$ for all $i \in \mathcal{I}$. Because the M_i function is a utility function and represents preferences, it follows that $M_i(c_i, l_i) \geq M_i(c_j, l_j)$ for all $i \in \mathcal{I}$. Hence, we have proven that

$$\min_{i \in \mathcal{I}} M_i(c_i, l_i) = M_j(c_j, l_j) = \min \left\{ \underline{h} + D - \tilde{h}; -\tau(0) - \tilde{w} \right\}$$

[33] Formally, for any l, it must be that $c \geq a(l)[-\tau(0)] + (1 - a(l))[\underline{h} + D]$ where c is such that $u_{\underline{i}}(c, l) = u_{\underline{i}}(c^*, l^*)$ and (c^*, l^*) maximizes $\succsim_{\underline{i}}$ over $B(\tau, D, 0, \underline{h})$.

By Theorem 1, the result follows. ■

Theorem 2 is useful because it amounts to identifying a sufficient statistic for evaluating the desirability of any tax-benefit reform, even in a suboptimal world. It can be used to compare a reform scenario (τ, D) against the status quo tax-benefit system (τ', D'). For example, if the gap between the lowest consumption and the average production in the home sector $(\underline{h} + D - \tilde{h})$ is larger in absolute value than in the labor market $(-\tau(0) - \tilde{w})$, then a welfare-improving reform would consist in raising the inactivity benefit D. I come back to the usefulness of this theorem in the empirical section 5. Let me also note that one could be concerned that Hardworking Poor Existence is too strong because it leads to pathological preferences as represented in Figure 5. Fortunately, I show in Appendix B.1. that similar results can be obtained with much weaker assumptions: we only need to assume that the slope of the tax-benefit system is tangent to the indifference curves of some agents, which is necessarily true if there are no gaps in the income distribution.

What is the intuition behind this result? Recall that, if there were no inequalities in productive skills (w_i, h_i), everyone would have a budget $(\tilde{w}, \tilde{h})$ and the absence of redistribution would be optimal by **Responsibility**. Hence, the government assesses welfare by computing the gap between the current situation of an individual and what she would have consumed in the ideal situation where she would have had the average budget $(\tilde{w}, \tilde{h})$. Because of the *maximin* nature of the social objective, any extra dollar should *fully* go to the agent who has the largest gap. As argued above, the social welfare function requires us to focus on redistribution among actives and among inactives, but not between them. Hence, the government computes that gap in the labor market and compares it to the gap at home. Overall, Theorem 2 implies that reducing within-sector inequality is welfare-improving in the sector that has the largest level of inequality, and the sectoral inequality measure is simply the difference between the smallest consumption and the average productivity.

Let me now turn to the characterization of the optimal tax benefit-system. It is enough to concentrate on D^* for our purposes.

Theorem 3. *Under Hardworking Poor Existence, for $e \in E$ with $\underline{w} = 0$, if there exists an incentive-compatible allocation decentralized by (τ^*, D^*) that is optimal with respect to the equality of opportunity social welfare function, then*

$$D^* = \tilde{h} - \underline{h} - \tau^*(0) - \tilde{w}$$

The proof is immediate from Theorem 2. The optimal inactivity benefit follows a similar formula for economies with $\underline{w} > 0$[34] and is relegated to Appendix B.1.

[34]The main difference is that the worst-off in the labor market may be different from the ones earnings $-\tau(0)$. For example, they may work full time at the minimum wage.

This optimal inactivity benefit formula is interesting for several reasons. First, it gives us an idea of the consumption difference that should hold between actives and inactives. In other words, this theorem pinpoints the premium that active agents *fairly* deserve because of their participation in the labor market, if any, the magnitude of which must be pinned down by empirical moments of the economy. In particular, observe that it could be the case that $D^* = -\tau^*(0)$ and social assistance is given to both actives and inactive agents. This is optimal when the bottom and the average productivities are equal across sectors, i.e. $\underline{h} = 0 = \underline{w}$ and $\tilde{h} = \tilde{w}$. This suggests that the discontinuity in transfers between inactive and unemployed is not a built-in feature of the extensive labor supply model, but rather depends on the particular economy under consideration.

Second, this premium is increasing in the discrepancy between average productivity in the labor market and at home $(\tilde{w} - \tilde{h})$. This suggests that as the labor market becomes relatively more productive, the inactivity benefit should decrease for a given τ^*. This would come as no surprise if we had introduced a government budget constraint. What is peculiar, surprising, and new here is that this holds only because of fairness considerations, whatever the size of efficiency costs.

Third, this theorem also informs us on the complementarity/substitutability of the inactivity benefit with the traditional safety net programs. In particular, basic income should supplement, rather than crowd out, the standard safety net as they covary in identical rather than opposite directions: an increase in one dollar of optimal social assistance $-\tau^*(0)$ should be accompanied with an increase of one dollar in optimal inactivity benefit D^*, *ceteris paribus*. Intuitively, if the public finance constraint is marginally relaxed and the optimal social assistance increases by a dollar, then the gap between the worst consumption at home and the average wage shrinks by a dollar. As a result, the government increases the inactivity benefit by a dollar to shrink the gap between the lowest and average home production by the same magnitude.

Finally, while Theorem 3 characterizes the relative level of the safety net and the inactivity benefit at the optimum, it is silent about the joint absolute level. To sketch its design[35], consider a given economy e and the *Laissez-faire* policy $(\tau, D) = (0, 0)$. The government computes the gap between the worst consumption and the average production in the labor market and in the home sector using Theorem 2. In most cases, that gap is much greater in the former than in the latter sector. Hence, the government will start by directing some transfers to the worst-off in the labor market, up to the point where the gaps between the worst consumption and the average production are equalized across sectors, as embodied by Theorem 3. In turn, the Rawlsian government

[35]It is easy to prove that τ^* would have the same qualitative properties as Fleurbaey and Maniquet (2007), i.e. setting $\underline{w} - \tau(\underline{w})$ as high as possible with negative marginal tax rates over $[0, \underline{w}]$.

will pursue tax collection by setting taxes to the revenue-maximizing rate, i.e. the tip of the Laffer curve, and transfer all the proceeds through D^* and $-\tau^*(0)$. In other words, it will direct redistribution such that any dollar spent on D is matched with a dollar spent on the safety net, thereby setting their joint level as high as efficiency permits.

5 Empirical application

Whether introducing an inactivity benefit is a welfare-improving reform or an optimal policy is ultimately an empirical question, as one can notice from Theorem 2 and Theorem 3, respectively. Let me start with the analysis of welfare-improving reforms.[36] I therefore need to estimate parameters $-\tau(0), \underline{h}, \tilde{w}, \tilde{h}$ from equation (1) in Theorem 2.

I set the right-hand side equation (1) with the status-quo tax-benefit system such that $(-\tau'(0), D') = (-\tau'(0), 0)$ for all countries as to date no OECD government has waived the conditionality of welfare benefits to labor market participation. Then, I calibrate the left handside with the reform that consists in giving a positive inactivity benefit, i.e. the reform $(-\tau(0), D)$ with $D > 0$. By Theorem 2, this reform is welfare-improving if

$$-\tau(0) - \tilde{w} > \underline{h} - \tilde{h} + D \iff -\tau(0) > \underline{h} - \tilde{h} + D + \tilde{w}$$

The right-hand side of this inequality reads as a sufficient statistic for the desirability of the reform of the inactivity benefit. In what follows, I consider a dollar increase of D (i.e. from 0 dollar to 1 dollar) and I compute the right-hand side. I obtain a sufficient statistic that answers the following question: *by how much should one increase the safety net in order for 1 dollar of inactivity benefit to be welfare-improving?* Hence, this section performs a bounding exercise.

The advantage of the sufficient statistics (1) is that estimates for the current social assistance $-\tau'(0)$ as well as the average gross earnings $\tilde{w}$ are readily available statistics for many countries[37]. However, the difficulty of this exercise is that home production surpluses estimates for $\underline{h}$ and $\tilde{h}$ are not readily available. From section 2, we know that h_i must be measured as the product of the sunk time cost of participating in the labor market[38] with the productivity in the home sector. Let me denote the former by F and

[36] Observed tax-benefit systems around the world are probably far away from the Mirrleesian optimum such that policy recommendations of practical use are more likely to emerge from the study welfare-improving reforms

[37] I recover $-\tau'(0)$ and $\tilde{w}$ from the OECD (2020) tax-benefit simulator for 29 developed economies. The results are differentiated for two different family compositions (i.e. two different status-quo τ'): childless singles and lone parents with two children. The reference year is set to 2020 for the case $\underline{w} = 0$ because there is little doubt that there has been involuntary unemployment.

[38] Observe that in the real world, this sunk time cost can only be partly controlled by the government (through job-seeking ordeals). This paper focuses on the redistribution problem considering that these ordeals are set for other reasons (see e.g. Rafkin et al. (2024)). Accordingly, the measurement of F is set to match commuting time, which is largely independent of these ordeals.

the latter by γ_i:

$$\underbrace{h_i}_{\text{home surplus}} = \overbrace{F}^{\text{Time cost of participation}} \times \underbrace{\gamma_i}_{\text{home hourly productivity}}$$

To estimate these two key unobservables, I will take a Beckerian view on home production and set average productivities to be identical across the two sectors, i.e. $\tilde{\gamma} = \tilde{w}$ (Becker, 1965). Moreover, I impose that $\underline{\gamma} = 0$. Observe that I only impose two moments restrictions, i.e. on the minimum and the average, while staying completely agnostic about the shape of the γ_i distribution. Section 5.1 addresses robustness.

To get an estimate of F, I exploit the recent G-SWA survey on time savings when working from home (Aksoy et al., 2023). On average across countries, workers spent 72 minutes per day commuting which is taken to reflect the sunk cost of labor market participation[39]. In turn, F is expressed as the fraction of this time cost over the statutory length of the working week[40] because l is normalized to 1. For example, the average American spends 55 minutes commuting per day over a 40 hours workweek, yielding a $F_{US} = 11.56\%$.

The results are reported in table 1.

Despite a series of conservative assumptions, I find that for all 29 countries, the safety net should be increased by very large amounts before any dollar spent on inactivity benefit constitutes a welfare improvement. The magnitude of estimates are in general smaller for lone parents than for childless singles, as most countries typically offer more generous coverage to the former than the latter. These results come from the fact that distances to the average are much larger in the labor market than in the home sector, or equivalently, that the well-being measurement pinned down by the axioms always identify the worst-off as being a job-seeker in these economies, but never an inactive. It is worth pointing out that, while results for the case $\underline{w} > 0$ are qualitatively similar, the magnitude is considerably smaller (see Appendix B.2). Yet, the policy conclusion is unchanged.

This analysis of welfare-improving reforms suggests that before introducing a basic income, a prioritarian government should significantly (most of the time, unrealistically) increase the safety net coverage offered to the actives. This comes from the fact that the empirical difference between inactives consuming $\underline{h}$ and unemployed consuming $-\tau(0)$ is smaller than $\tilde{h} - \tilde{w}$.

[39]An alternative strategy would have been to use estimates of the time devoted to job search activities. However, Mukoyama et al. (2018) documented that unemployed Americans spend on average 31.1 minutes per day searching for a job, such that my choice is again conservative.

[40]Additional details on the empirical application are relegated to Appendix D.

Table 1: Summary of results. Sufficient increase in $-\tau(0)$ for $D = 1$ to be a welfare-improving reform, in percentage of current $-\tau(0)$.

Country	Lone parents	Singles
Australia	365.99%	410.22%
Belgium	142.67%	248.18%
Bulgaria	352.10%	1109.41%
Canada	353.95%	600.39%
Czech Republic	303.34%	730.34%
Estonia	116.44%	657.54%
France	121.11%	371.55%
Greece	294.24%	589.93%
Germany	218.40%	744.30%
Croatia	325.47%	793.48%
Hungary	1490.03%	1490.03%
Israel	223.35%	549.81%
Ireland	186.29%	288.84%
Japan	68.66%	319.12%
Lithuania	224.54%	788.99%
Latvia	302.19%	1307.68%
Luxembourg	82.64%	189.04%
Malta	205.47%	235.06%
Netherlands	257.60%	257.60%
New Zealand	164.09%	295.15%
Poland	188.73%	994.14%
Portugal	244.71%	589.42%
Romania	1012.90%	2697.91%
Slovenia	35.97%	245.36%
Slovak Republic	511.34%	1365.17%
Spain	220.58%	374.47%
Turkey	3639.13%	∞
United Kingdom	226.69%	638.80%
United States	762.76%	2163.63%

Now, recall that our previous theorems not only informed us of the direction of welfare-improving reforms but also on the design of optimal redistribution. Theorem 2 gives a sufficient statistic that estimate by how much we should increase social assistance before a dollar of basic income is welfare-improving, while Theorem 3 says how high should basic income be if the current social assistance is assumed to be optimal.[41] Conveniently, these two exercises are numerically equivalent, in the sense that they deliver the same estimates (i.e. the estimates of Table 1). For example, if one assumes that the observed Luxembourgian social assistance programs are optimal $(-\tau'(0) = -\tau^*(0))$, then the optimal inactivity benefit for lone parents in the Luxembourg should amount to -82.64% multiplied by the level of social assistance. In other words, all optimal inactivity benefits should be negative. Overall, this goes against the idea that introducing a basic income without modifying the safety net would be welfare-improving, let alone optimal.

5.1 Discussion

I made several assumptions that, if relaxed, would render the conflict between basic income and equality of opportunity even stronger. I discuss them in turn for the empirical application, the theoretical framework and the government's objective.

From the theorems of section 4, we know that a positive basic income is more likely to emerge if the gap $\tilde{w} - \tilde{h}$ is small. In order for my negative result on its desirability to be robust, I have made measurement assumptions that were favorable to the emergence of a basic income, i.e. conservative assumptions. First, the ratio of the average productivity in the home sector to the labor market $\frac{\tilde{\gamma}}{\tilde{w}}$ has been set to 1 while Bridgman et al. (2018) found it closer to 0.3. Second, the choice of F could have been smaller if it had reflected time devoted to job search rather than commuting (see Mukoyama et al. (2018)). Third, I assumed that the average marginal product of labor $\tilde{w}$ is equal to the average gross earnings. However, as firms have monopsony power, the marginal product of labor is higher than the wage[42]. All in all, these choices suggest that estimates in table 1 are lower bounds.

Moreover, several features of the theoretical framework are also conservative with respect to this conflict. First, I considered that there is no negative externality of home production. However, if it includes black market activities, the government might wish to impose a Pigouvian tax on inactives and/or hold them responsible for their h_i, thereby decreasing even more the desirable level of D^*. Second, the model assumed away the

[41]This interpretation of the results follows the line of the inverse-optimum. See Bourguignon and Spadaro (2012) and Stantcheva (2016) for an introduction and a critique of the inverse-optimum approach, respectively.

[42]Azar and Marinescu (2024) and Mas and Pallais (2019) review the literature and consider that the marginal product of labor may be 25% larger than the average wage.

existence of an intensive margin in the inactives' production function. As in Saez (2001, 2002) or Rothschild and Scheuer (2013), such an intensive margin would have driven an additional efficiency cost of raising D by disincentivizing effort in the home sector. Hence, these results would only be reinforced by including an intensive margin.

However, I have assumed that the government can perfectly distinguish a job-seeker from an inactive at a zero cost, which might seem a strong assumption. Yet, as long as the monitoring cost for the government has a lower dollar value than estimates from Table 1, the government should fully invest in it. Given the size of the estimates, the inclusion of costly monitoring is unlikely to overturn the policy recommendation. In the same vein, if the screening mechanism is imperfect and a basic income emerges as optimal policy, then this policy would be justified on third-best considerations but not on second-best arguments (see Ghatak and Maniquet (2019) for a discussion).

Finally, this result holds despite the fact that the social objective have not favored the production in the formal sector with respect to the one in the home sector[43] even if there may be good reasons to do so. Indeed, there have been recent calls in the public debate for an increase in the employment rate, notably in order to maintain the sustainability of public pension systems in aging economies.

However, this paper did not exclude the possibility that other social objectives may lead to different policy recommendations. In particular, among the family of social objectives pursuing equality of opportunity, the one studied above may be criticized by some because it holds agents responsible for their disutilities of participation. In Appendix C, I study the case where disutilities of participation are driven by a welfare recipient stigma[44] that the government wishes to compensate for. It is shown that the estimates in Table 1 can be then interpreted as lower bounds for the government's willingness to compensate for the stigma cost of conditionality.

6 Conclusion

This article has explored redistribution between active and inactive agents. The inquiry showed that abandoning the conditionality of social benefits on labor market participation is unlikely to be justified by the ethics of equality of opportunity in most developed

[43]This is especially relevant to the debate because proponents of basic income have argued that one should not be paternalistic about what a good life is (see in particular Van Parijs (1995)).

[44]This *welfare stigma hypothesis* has received attention from the theoretical literature (Besley & Coate, 1992a; Hupkau & Maniquet, 2018; Lindbeck et al., 1999; Moffitt, 1983) and was recently backed by experimental evidence (Friedrichsen et al., 2018).

economies.[45] Equivalently, if one wants to introduce a basic income in these economies, it can only be justified by a social welfare function that encourages redistribution across preferences.

The main explanation lies in the fact that as the value of home production is small, inactivity is driven by preferences for which agents are held responsible. Hence, an inequality-averse government can fight inequalities outside the labor force by providing better opportunities within the labor market, which is desirable under the proviso that the aggregate technology in the formal sector is productive enough.

This paper assumed a perfect observability of the activity status a_i which might be noisy in practice.[46] This could justify the emergence of a basic income for third-best considerations while the present paper is confined to the analysis of the second-best. Similarly, this paper has left exogenous the definition of the eligibility requirements to be considered as an active job-seeker. However, it is well-known that there exists a heterogeneity in the stringency of these requirements across developed countries, whose positive and normative study is left for future research.

[45] A priori, this ethical standpoint could have justified both an anti- and a pro-basic income argument as could be attested by the debate between Rawls (1988) and Van Parijs (1991) on whether Malibu surfers should be fed. What the present paper has done is precisely to reconcile these two diametrically opposed interpretation of a single fairness viewpoint, by proving the conjecture outlined in Van Parijs and Vanderborght (2017, p.112): "*[Once leisure is included in the index], the optimal option, by the standard of the difference principle, will crucially depend on the relative weights the index places on income and leisure, [...] and on a great many contingent empirical facts*". In this paper, the relative weights on these dimensions are those of the agents themselves because of the non-paternalistic nature of the social objective. Then I quantify where and when these empirical facts are such that a basic income may not be justified: in developed economies.

[46] Rafkin et al. (2024) study the correlations between eligibility and take-up for various US transfer programs. They find that the take-up is concentrated among eligible individuals with the worst consumption levels. Hence, the correlation is positive but below 1.

References

Aksoy, C. G., Barrero, J. M., Bloom, N., Davis, S. J., Dolls, M., & Zarate, P. (2023). Time savings when working from home. *National Bureau of Economic Research Working Papers*.

Arrow, K. J. (1950). A difficulty in the concept of social welfare. *Journal of Political Economy*, *58*(4), 328–346.

Atkinson, A. B. (1996). *Public economics in action: The basic income/flat tax proposal*. Oxford University Press.

Azar, J., & Marinescu, I. (2024). Monopsony power in the labor market. *Handbook of labor economics* (pp. 761–827). Elsevier.

Banerjee, A., Niehaus, P., & Suri, T. (2019). Universal basic income in the developing world. *Annual Review of Economics*, *11*, 959–983.

Baqaee, D. R., Burstein, A. T., & Koike-Mori, Y. (2024). Measuring welfare by matching households across time. *The Quarterly Journal of Economics*, *139*(1), 533–573.

Beaudry, P., Blackorby, C., & Szalay, D. (2009). Taxes and employment subsidies in optimal redistribution programs. *American Economic Review*, *99*(1), 216–242.

Becker, G. S. (1965). A theory of the allocation of time. *The Economic Journal*, *75*(299), 493–517.

Bernheim, B. D. (2009). Behavioral welfare economics. *Journal of the European Economic Association*, *7*(2-3), 267–319.

Bernheim, B. D., & Taubinsky, D. (2018). Behavioral public economics. *Handbook of behavioral economics: Applications and Foundations 1*, *1*, 381–516.

Besley, T., & Coate, S. (1992a). Understanding welfare stigma: Taxpayer resentment and statistical discrimination. *Journal of Public Economics*, *48*(2), 165–183.

Besley, T., & Coate, S. (1992b). Workfare versus welfare: Incentive arguments for work requirements in poverty-alleviation programs. *American Economic Review*, *82*(1), 249–261.

Besley, T., & Coate, S. (1995). The design of income maintenance programmes. *The Review of Economic Studies*, *62*(2), 187–221.

Bierbrauer, F., Boyer, P. C., & Peichl, A. (2021). Politically feasible reforms of nonlinear tax systems. *American Economic Review*, *111*(1), 153–91.

Boadway, R., & Cuff, K. (2014). Monitoring and optimal income taxation with involuntary unemployment. *Annals of Economics and Statistics/Annales d'Économie et de Statistique*, (113/114), 121–157.

Boadway, R., & Cuff, K. (2018). Optimal unemployment insurance and redistribution. *Journal of Public Economic Theory*, *20*(3), 303–324.

Boadway, R., Cuff, K., & Marceau, N. (2003). Redistribution and employment policies with endogenous unemployment. *Journal of Public Economics*, *87*(11), 2407–2430.

Boerma, J., & Karabarbounis, L. (2021). Inferring inequality with home production. *Econometrica*, *89*(5), 2517–2556.

Boone, J., & Bovenberg, L. (2013). Optimal taxation and welfare benefits with monitoring of job search. *International Tax and Public Finance*, *20*(2), 268–292.

Bosmans, K., Decancq, K., & Ooghe, E. (2018). Who's afraid of aggregating money metrics? *Theoretical Economics*, *13*(2), 467–484.

Bourguignon, F., & Spadaro, A. (2012). Tax–benefit revealed social preferences. *The Journal of Economic Inequality*, *10*(1), 75–108.

Bridgman, B., Duernecker, G., & Herrendorf, B. (2018). Structural transformation, marketization, and household production around the world. *Journal of Development Economics*, *133*, 102–126.

Cesarini, D., Lindqvist, E., Notowidigdo, M. J., & Östling, R. (2017). The effect of wealth on individual and household labor supply: Evidence from swedish lotteries. *American Economic Review*, *107*(12), 3917–3946.

Choné, P., & Laroque, G. (2005). Optimal incentives for labor force participation. *Journal of Public Economics*, *89*(2-3), 395–425.

Choné, P., & Laroque, G. (2011). Optimal taxation in the extensive model. *Journal of Economic Theory*, *146*(2), 425–453.

Conesa, J. C., Li, B., & Li, Q. (2023). A quantitative evaluation of universal basic income. *Journal of Public Economics*, *223*, 104881.

Daruich, D., & Fernández, R. (2024). Universal basic income: A dynamic assessment. *American Economic Review*, *114*(1), 38–88.

Diamond, P. (1980). Income taxation with fixed hours of work. *Journal of Public Economics*, *13*(1), 101–110.

Dworczak, P. et al. (2023). *Equity-efficiency trade-off in quasi-linear environments* (tech. rep.). Working paper.

Dworkin, R. (1981). What is equality? part 2: Equality of resources. *Philosophy & Public Affairs*, 283–345.

Dworkin, R. (2000). *Sovereign virtue: The theory and practice of equality*. Harvard University Press.

Edgeworth, F. Y. (1897). The pure theory of taxation. *The Economic Journal*, *7*(25), 46–70.

Ferey, A. (2022). Redistribution and unemployment insurance.

Fleming, M. (1952). A cardinal concept of welfare. *The Quarterly Journal of Economics*, *66*(3), 366–384.

Fleurbaey, M. (2008). *Fairness, responsibility, and welfare*. Oxford University Press.

Fleurbaey, M., & Maniquet, F. (2006). Fair income tax. *The Review of Economic Studies*, *73*(1), 55–83.

Fleurbaey, M., & Maniquet, F. (2007). Help the low skilled or let the hardworking thrive? a study of fairness in optimal income taxation. *Journal of Public Economic Theory*, *9*(3), 467–500.

Fleurbaey, M., & Maniquet, F. (2011). *A theory of fairness and social welfare*. Econometric Society Monograph (vol 48). Cambridge University Press.

Fleurbaey, M., & Maniquet, F. (2017). Fairness and well-being measurement. *Mathematical Social Sciences*, *90*, 119–126.

Fleurbaey, M., & Maniquet, F. (2018). Optimal income taxation theory and principles of fairness. *Journal of Economic Literature*, *56*(3), 1029–79.

Friedrichsen, J., König, T., & Schmacker, R. (2018). Social image concerns and welfare take-up. *Journal of Public Economics*, *168*, 174–192.

Gayle, G.-L., & Shephard, A. (2019). Optimal taxation, marriage, home production, and family labor supply. *Econometrica*, *87*(1), 291–326.

Ghatak, M., & Maniquet, F. (2019). Universal basic income: Some theoretical aspects. *Annual Review of Economics*, *11*, 895–928.

Golosov, M., Graber, M., Mogstad, M., & Novgorodsky, D. (2024). How americans respond to idiosyncratic and exogenous changes in household wealth and unearned income. *Quaterly Journal of Economics*.

Hansen, E. (2021). Optimal income taxation with labor supply responses at two margins: When is an earned income tax credit optimal? *Journal of Public Economics*, *195*, 104365.

Hansson, B. (1973). The independence condition in the theory of social choice. *Theory and Decision*, *4*(1), 25–49.

Hoynes, H., & Rothstein, J. (2019). Universal basic income in the united states and advanced countries. *Annual Review of Economics*, *11*, 929–958.

Hungerbühler, M., & Lehmann, E. (2009). On the optimality of a minimum wage: New insights from optimal tax theory. *Journal of Public Economics*, *93*(3-4), 464–481.

Hungerbühler, M., Lehmann, E., Parmentier, A., & Van der Linden, B. (2006). Optimal redistributive taxation in a search equilibrium model. *The Review of Economic Studies*, *73*(3), 743–767.

Hupkau, C., & Maniquet, F. (2018). Identity, non-take-up and welfare conditionality. *Journal of Economic Behavior & Organization*, *147*, 13–27.

Jacquet, L., Lehmann, E., & Van der Linden, B. (2013). Optimal redistributive taxation with both extensive and intensive responses. *Journal of Economic Theory*, *148*(5), 1770–1805.

Jacquet, L., Lehmann, E., & Van der Linden, B. (2014). Optimal income taxation with kalai wage bargaining and endogenous participation. *Social Choice and Welfare*, *42*(2), 381–402.

Jaravel, X., & Lashkari, D. (2024). Measuring growth in consumer welfare with income-dependent preferences: Nonparametric methods and estimates for the united states. *The Quarterly Journal of Economics*, *139*(1), 477–532.

Kaplan, G., & Schulhofer-Wohl, S. (2018). The changing (dis-) utility of work. *Journal of Economic Perspectives*, *32*(3), 239–58.

Kleven, H. J., & Kopczuk, W. (2011). Transfer program complexity and the take-up of social benefits. *American Economic Journal: Economic Policy*, *3*(1), 54–90.

Kroft, K., Kucko, K., Lehmann, E., & Schmieder, J. (2020). Optimal income taxation with unemployment and wage responses: A sufficient statistics approach. *American Economic Journal: Economic Policy*, *12*(1), 254–92.

Lehmann, E., Parmentier, A., & Van der Linden, B. (2011). Optimal income taxation with endogenous participation and search unemployment. *Journal of Public Economics*, *95*(11-12), 1523–1537.

Lindbeck, A., Nyberg, S., & Weibull, J. W. (1999). Social norms and economic incentives in the welfare state. *The Quarterly Journal of Economics*, *114*(1), 1–35.

Mas, A., & Pallais, A. (2019). Labor supply and the value of non-work time: Experimental estimates from the field. *American Economic Review: Insights*, *1*(1), 111–126.

Mirrlees, J. A. (1971). An exploration in the theory of optimum income taxation. *The Review of Economic Studies*, *38*(2), 175–208.

MISSOC. (2021). Mutual information system on social protection database [data retrieved from https://www.missoc.org/missoc-database/comparative-tables/].

Moffitt, R. (1983). An economic model of welfare stigma. *American Economic Review*, *73*(5), 1023–1035.

Mukoyama, T., Patterson, C., & Şahin, A. (2018). Job search behavior over the business cycle. *American Economic Journal: Macroeconomics*, *10*(1), 190–215.

OECD. (2020). *TaxBEN: The OECD tax-benefit simulation model*. OECD Publishing.

Piacquadio, P. G. (2017). A fairness justification of utilitarianism. *Econometrica*, *85*(4), 1261–1276.

Rafkin, C., Solomon, A., & Soltas, E. (2024). Self-targeting in us transfer programs. *Working Paper SSRN 4495537*.

Rawls, J. (1971). *A theory of justice*. Belknap Press.

Rawls, J. (1988). The priority of right and ideas of the good. *Philosophy & Public Affairs*, 251–276.

Rothschild, C., & Scheuer, F. (2013). Redistributive taxation in the roy model. *The Quarterly Journal of Economics*, *128*(2), 623–668.

Roy, A. D. (1951). Some thoughts on the distribution of earnings. *Oxford economic papers*, *3*(2), 135–146.

Saez, E. (2001). Using elasticities to derive optimal income tax rates. *The Review of Economic Studies*, *68*(1), 205–229.

Saez, E. (2002). Optimal income transfer programs: Intensive versus extensive labor supply responses. *The Quarterly Journal of Economics*, *117*(3), 1039–1073.

Saez, E., & Stantcheva, S. (2016). Generalized social marginal welfare weights for optimal tax theory. *American Economic Review*, *106*(1), 24–45.

Samuelson, P. A., & Swamy, S. (1974). Invariant economic index numbers and canonical duality: Survey and synthesis. *American Economic Review*, *64*(4), 566–593.

Sher, I. (2024). Generalized social marginal welfare weights imply inconsistent comparisons of tax policies. *American Economic Review*, *114*(11), 3551–3577.

Stantcheva, S. (2016). Comment on "positive and normative judgments implicit in us tax policy and the costs of unequal growth and recessions". *Journal of Monetary Economics*, *100*(77), 48–52.

Stantcheva, S. (2021). Understanding tax policy: How do people reason? *The Quarterly Journal of Economics*, *136*(4), 2309–2369.

Van Parijs, P. (1991). Why surfers should be fed: The liberal case for an unconditional basic income. *Philosophy and Public Affairs*, *20*(2), 101.

Van Parijs, P. (1995). *Real freedom for all: What (if anything) can justify capitalism?* Clarendon Press.

Van Parijs, P. (2021). Unjust inequalities. is maximin the answer? *In A. Deaton (Ed.) The Institute for Fiscal Studies Deaton Review*.

Van Parijs, P., & Vanderborght, Y. (2017). *Basic income: A radical proposal for a free society and a sane economy*. Harvard University Press.

Weinzierl, M. (2017). Popular acceptance of inequality due to innate brute luck and support for classical benefit-based taxation. *Journal of Public Economics*, *155*, 54–63.

A Axiomatic proofs

A.1 Axioms

In this section, I define formally each axiom used in the paper. Let me define the social ordering function (SOF) $\succsim$ for an economy $e \in E$ as a transitive ordering such that for any two allocations $(c, l), (c', l') \in X^{\mathcal{I}}$, such that we write $(c, l) \succsim (c', l')$ whenever (c, l) is socially weakly preferred to (c', l'). The strict social preference and the social indifference are denoted by $\succ$ and $\sim$, respectively.

The first axiom is the celebrated Pareto principle, which states that the planner must agree with unanimous decisions.

Axiom 1: Weak Pareto
If $(c_i, l_i) \succ_i (c'_i, l'_i)$ for all $i \in \mathcal{I}$ then $(c, l) \succ (c', l')$.

The second axiom holds agents responsible for inequalities spawned by their preferences. It consists in saying that when all agents have identical endowments, reducing lump-sum transfers inequality is desirable. In the limit, Laissez-faire is optimal in this knife-hedge case where there is no inequality in endowments. It is illustrated in Figure A.1.

Axiom 2: Responsibility
For all economy $e \in E$ and allocations with $(w_i, h_i) = (w_0, h_0)\ \forall i \in \mathcal{I}$,
If $\exists i, j \in \mathcal{I}$ and $(c, l), (c', l') \in X^{\mathcal{I}}$ with

$$(c_i, l_i) \in \max_{\succsim_i} B(t_i, w_0, h_0) \quad (c'_i, l'_i) \in \max_{\succsim_i} B(t'_i, w_0, h_0)$$
$$(c_j, l_j) \in \max_{\succsim_j} B(t_j, w_0, h_0) \quad (c'_j, l'_j) \in \max_{\succsim_j} B(t'_j, w_0, h_0)$$

and $(c_k, l_k) = (c'_k, l'_k)$ for all $k \in \mathcal{I} \backslash \{i, j\}$ and $\exists \delta > 0$ such that $t'_i - \delta = t_i \geq t_j = t'_j + \delta$, then $(c, l) \succ (c', l')$.

The third axiom compensates agents for unqual skills (w_i, h_i). It is illustrated in FIgure A.2.

Axiom 3: Weak Transfer
For all economy $e \in E$, all allocations $(c, l), (c', l') \in X^{\mathcal{I}}$, if $\exists i, j \in \mathcal{I}$ two agents with $\succsim_i = \succsim_j$ such that

$$l_i = l_j = l'_i = l'_j$$

and for some $\delta > 0$

$$c'_i - \delta = c_i \geq c_j = c'_j + \delta$$

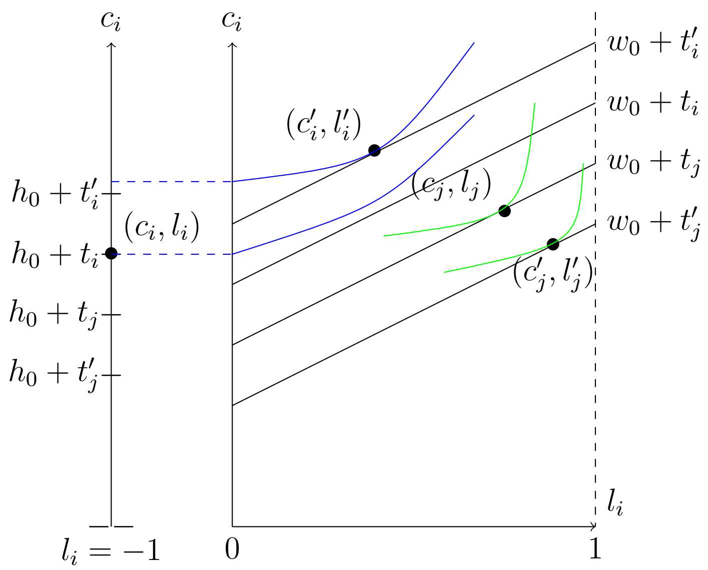

Figure A.1: In this figure, all agents have the same endowments w_i and h_i. Yet, they have received different lump-sum transfers for some reasons. **Responsibility** imposes that $(c, l) \succ (c', l')$ because lump-sum transfer inequality is reduced in (c, l).

while $(c_k, l_k) = (c'_k, l'_k)\ \forall k \in \mathcal{I} \backslash \{i, j\}$;
Then, $(c, l) \succsim (c', l')$

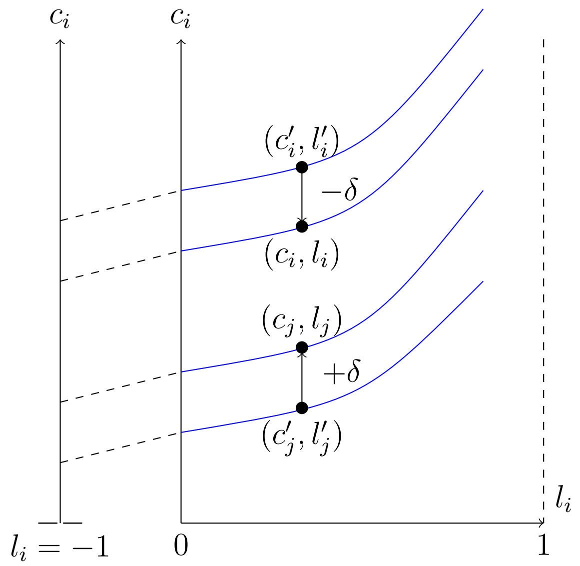

Figure A.2: **Weak Transfer** imposes that $(c, l) \succsim (c', l')$

Now, in order to build a SOF *for all economies* one needs consistency conditions i.e. invariance rules of the social evaluation when the economy changes. A popular choice in the literature is *Separability* which prescribes that adding or removing from the economy indifferent agents should not affect the ranking between two allocations.

Axiom: Separability
If there exists $(c,l),(c',l') \in X^{\mathcal{I}}$ such that $(c_j,l_j)=(c'_j,l'_j)$ then

$$(c,l) \succsim (c',l') \iff (c,l)_{(\mathcal{I}\backslash\{j\})} \succsim_{(\mathcal{I}\backslash\{j\})} (c',l')_{(\mathcal{I}\backslash\{j\})}$$

with $(c,l)_{(\mathcal{I}\backslash\{j\})} = \Big((c_1,l_1),...(c_{j-1},l_{j-1}),(c_{j+1},l_{j+1}),...,(c_I,l_I)\Big)$
and $(c',l')_{(\mathcal{I}\backslash\{j\})} = \Big((c'_1,l'_1),...(c'_{j-1},l'_{j-1}),(c'_{j+1},l'_{j+1}),...,(c'_I,l'_I)\Big)$ and $\succsim_{(\mathcal{I}\backslash\{j\})}$ is the SOF when the set of agents is $\mathcal{I}\backslash\{j\}$

We will also use a logical weakening of the latter, i.e. ***Mean-Preserving Separability*** which prescribes that removing indifferent agents from the economy does not affect the social ranking only if the absence of these agents do not change the per-capita amount of resources.

Axiom 4: Mean-Preserving Separability
For all $\mathcal{S} \subset \mathcal{I}$ and for all economy $e, e_{(\mathcal{I}\backslash\mathcal{S})} \in E$ with $e_{(\mathcal{I}\backslash\mathcal{S})} = \Big((w_i,h_i,\succsim_i)_{\forall i\in\mathcal{I}\backslash\mathcal{S}}\Big)$, and for all $(c,l),(c',l') \in X^{\mathcal{I}}$ such that

- $(c_i,l_i)=(c'_i,l'_i)$ for all $i \in \mathcal{S}$ and
- $\frac{1}{I}\sum_{i=1}^{I} w_i = \frac{1}{S}\sum_{i=1}^{S} w_i$ and $\frac{1}{I}\sum_{i=1}^{I} h_i = \frac{1}{S}\sum_{i=1}^{S} h_i$

Then $(c,l) \succsim (c',l') \iff (c,l)_{(\mathcal{I}\backslash\mathcal{S})} \succsim_{(\mathcal{I}\backslash\mathcal{S})} (c',l')_{(\mathcal{I}\backslash\mathcal{S})}$
where $(c,l)_{(\mathcal{I}\backslash\mathcal{S})},(c',l')_{(\mathcal{I}\backslash\mathcal{S})} \in X^{\mathcal{I}\backslash\mathcal{S}}$ and $\succsim_{(\mathcal{I}\backslash\mathcal{S})}$ is the social ordering over the subpopulation $\mathcal{I}\backslash\mathcal{S}$.

Finally, the fifth axiom, ***Hansson (1973) Independence***, deals with the informational structure of the SOF. It weakens the Arrovian binary independence in order to escape the impossibility of social choice. It imposes that when the indifference curves over two allocations are unchanged between two economies, then the social ordering over these allocations is unchanged as well.

Axiom 5: Hansson (1973) independence
For all economy $e = \Big((w_i,h_i,\succsim_i)_{\forall i\in I}\Big)$ $e' = \Big((w_i,h_i,\succsim'_i)_{\forall i\in I}\Big) \in E$ with $(\succsim_i)_{i\in\mathcal{I}}$ and $(\succsim'_i)_{i\in\mathcal{I}}$ two profiles of preferences, let $(c,l),(c',l') \in X^{\mathcal{I}}$ be two allocations,

$$\text{If } \forall q \in X \ \Big[(c_i,l_i) \sim_i q \iff (c_i,l_i) \sim'_i q \text{ and } (c'_i,l'_i) \sim_i q \iff (c'_i,l'_i) \sim'_i q \ \forall i \in \mathcal{I}\Big]$$
$$\text{Then, } [z \succsim z' \iff z \succsim' z']$$

where $\succsim'$ is the SOF when the economy is e'.

A.2 Theorems and proofs

I now prove that the standard axiom of *Separability* clashes with **Responsibility** and **Weak Transfer**. To the best of my knowledge, this result is new. The key difference with previous papers is that the present paper has three dimensions of heterogeneity $(\succsim_i, w_i, h_i)$ while most previous papers only had two (Fleurbaey & Maniquet, 2018).

Theorem A.1. *There is no SOF that satisfies* ***Responsibility, Weak Transfer*** *and Separability for all* $e \in E$

Proof. By contradiction, suppose the statement does not hold. Consider the economies $e_1 = \{(\succsim_i, \bar{h}, \underline{w}), (\succsim_j, \bar{h}, \underline{w})\}$, $e_2 = \{(\succsim_i, \underline{h}, \bar{w}), (\succsim_j, \underline{h}, \bar{w})\}$, $e_3 = \{(\succsim_i, \bar{h}, \underline{w}), (\succsim_j, \bar{h}, \underline{w})(\succsim_i, \underline{h}, \bar{w}), (\succsim_j, \underline{h}, \bar{w})\}$, with $\bar{h} > \underline{h}$ and $\bar{w} > \underline{w}$, as well as their associated utility-maximizing bundles:

$$(c_1, l_1) \in \max_{\succsim_i} B(0, \bar{h}, \underline{w}) \qquad (c_2, l_2) \in \max_{\succsim_i} B(0, \underline{h}, \bar{w})$$
$$(c_3, l_3) \in \max_{\succsim_j} B(0, \bar{h}, \underline{w}) \qquad (c_4, l_4) \in \max_{\succsim_j} B(0, \underline{h}, \bar{w})$$

The economies have been constructed such that $c_3 - c_4 = c_1 - c_2$, and $l_1 = l_2$ as well

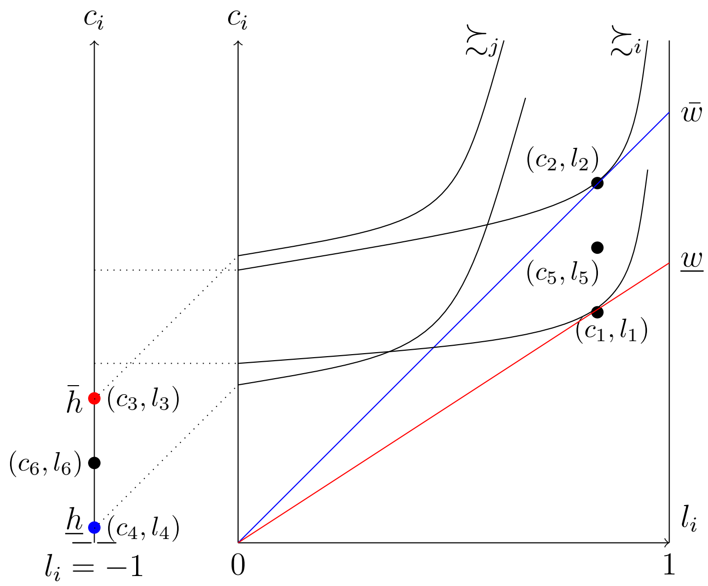

Figure A.3: Illustration of the proof of Theorem A.1.

$l_3 = l_4$. I define the average bundles $(c_5, l_5) = \frac{(c_1,l_1)+(c_2,l_2)}{2}$ and $(c_6, l_6) = \frac{(c_3,l_3)+(c_4,l_4)}{2}$ and illustrate the setup in Figure A.3.. It is enough to prove that the SOF leads to a contradiction in these particular economies for the statement to hold.

By **Responsibility**, $\Big((c_1, l_1), (c_3, l_3)\Big) \succ_{(e_1)} \Big((c_5, l_5), (c_6, l_6)\Big)$,

By **Separability**, $\Big((c_1, l_1), (c_3, l_3), (c_2, l_2), (c_4, l_4)\Big) \succ_{(e_3)} \Big((c_5, l_5), (c_6, l_6), (c_2, l_2), (c_4, l_4)\Big)$,

By **Weak Transfer**, $\Big((c_5, l_5), (c_6, l_6), (c_5, l_5), (c_6, l_6)\Big) \succsim_{(e_3)} \Big((c_1, l_1), (c_3, l_3), (c_2, l_2), (c_4, l_4)\Big)$,

By **Transitivity**, $\Big((c_5,l_5),(c_6,l_6),(c_5,l_5),(c_6,l_6)\Big) \succ_{(e3)} \Big((c_5,l_5),(c_6,l_6),(c_2,l_2),(c_4,l_4)\Big)$,
By **Separability**, $\Big((c_5,l_5),(c_6,l_6)\Big) \succ_{(e2)} \Big((c_2,l_2),(c_4,l_4)\Big)$,

But this contradicts **Responsibility**, proving the statement. ■

The main-text Theorem 1 shows that one can escape the impossibility of Theorem A.1 by weakening **Separability** to **Mean-Preserving Separability.** I prove Theorem 1 by using two lemmas. This proof is reminiscent to Fleurbaey and Maniquet (2006, 2011) and Valletta (2014)[47]. Lemma 1 implies the maximin[48] nature of the social ordering while Lemma 2 characterize the well-being measure.

Lemma 1. *If the SOF satisfies **Weak Pareto, Hansson independence and Weak Transfer,** then $\forall e \in E$, $(c,l),(c',l') \in X^{\mathcal{I}}$ if there exists $\{i,j\} \in \mathcal{I}$ such that $\succsim_i = \succsim_j \equiv \succsim_0$ and*

$$(c'_i,l'_i) \succ_0 (c_i,l_i) \succ_0 (c_j,l_j) \succ_0 (c'_j,l'_j)$$

while $(c'_k,l'_j)=(c_k,l_k)$ for all $k \in \mathcal{I}\backslash\{i,j\}$, one has $(c,l) \succ (c',l')$.

Proof of Lemma 1. Follows mutatis mutandis the proof of Lemma 1 in Fleurbaey and Maniquet (2006). ■

Lemma 2. *If the SOF satisfies **Weak Pareto, Responsibility, Weak Transfer, Hansson Independence, and Mean-Preserving Separability,** and $\forall e \in E$, $\exists (c,l),(c',l') \in X^{\mathcal{I}}$ such that*

$$M_i(c'_i,l'_i) > M_i(c_i,l_i) > M_j(c_j,l_j) > M_j(c'_j,l'_j)$$

and $(c_k,l_j)=(c'_k,l'_k)$ $\forall k \in \mathcal{I}\backslash\{i,j\}$
Then,

$$(c,l) \succ (c',l')$$

Proof of Lemma 2. By contradiction, suppose that $(c',l') \succsim (c,l)$.
Let me introduce two new agents, a,b such that:

- $(w_a,h_a)=(w_b,h_b)=(\tilde{w},\tilde{h})$
- $\succsim_a = \succsim_i$ and $\succsim_b = \succsim_j$

[47] In comparison with Fleurbaey and Maniquet (2006), I have a stronger Responsibility requirement but a weaker Separability requirement and more dimensions of heterogeneity. With respect to Valletta (2014), I have weaker versions of Pareto and Transfer axiom. That paper dealt with the fair income tax if there are two consumption goods but only one productive skill. In the present paper, I have an homogeneous consumption good but productive skills in two sectors.

[48] There has been recent axiomatizations of money-metric aggregator with finite inequality aversion which consists in weakening either **Weak Transfer** (Bosmans et al., 2018) or **Hansson Independence** (Piacquadio, 2017). However, I kept the present structure as the maximin has been crucial in the basic income debate.

I denote the relevant economies in the following way :

$$e_{(\{a,b\})} = \Big((w_i, h_i, \succsim_i)_{\forall i \in \{a,b\}} \Big) \qquad e_{(\mathcal{I} \cup \{a,b\})} = \Big((w_i, h_i, \succsim_i)_{\forall i \in \mathcal{I} \cup \{a,b\}} \Big)$$

Let $\Big((c_a, l_a), (c_b, l_b) \Big), \Big((c'_a, l'_a), (c'_b, l'_b) \Big)$ be two allocations in $e^{\{a,b\}}$ such that

$$(c_a, l_a) \in \max_{\succsim_a} B(t_a, \tilde{w}, \tilde{h}) \quad (c'_a, l'_a) \in \max_{\succsim_a} B(t'_a, \tilde{w}, \tilde{h})$$
$$(c_b, l_b) \in \max_{\succsim_b} B(t_b, \tilde{w}, \tilde{h}) \quad (c'_b, l'_b) \in \max_{\succsim_b} B(t'_b, \tilde{w}, \tilde{h})$$

with $t_a > t'_a > t'_b > t_b$
and

$$M_i(c'_i, l'_i) > M_i(c_i, l_i) > M_a(c_a, l_a) > M_a(c'_a, l'_a) > M_b(c'_b, l'_b) > M_b(c_b, l_b) > M_j(c_j, l_j) > M_j(c'_j, l'_j)$$

By **Mean-Preserving Separability,**

$$\Big((c', l'), (c'_a, l'_a), (c'_b, l'_b) \Big) \succsim_{(\mathcal{I} \cup \{a,b\})} \Big((c, l), (c'_a, l'_a), (c'_b, l'_b) \Big)$$

Recall that $(c_k, l_k) = (c'_k, l'_k)$ for all $k \in I \backslash \{i, j\}$. Moreover, because $M_i(\cdot)$ and $M_a(\cdot)$ represents the same preferences (resp. $M_j(\cdot)$ and $M_b(\cdot)$), we can apply Lemma 1 between agents i and a (resp. between agents j and b) to get

$$\Big((c, l), (c_a, l_a), (c_b, l_b) \Big) \succ_{(\mathcal{I} \cup \{a,b\})} \Big((c', l'), (c'_a, l'_a), (c'_b, l'_b) \Big)$$

By transitivity of the SOF,

$$\Big((c, l), (c_a, l_a), (c_b, l_b) \Big) \succ_{(\mathcal{I} \cup \{a,b\})} \Big((c, l), (c'_a, l'_a), (c'_b, l'_b) \Big)$$

By **Mean-Preserving Separability** one has,

$$\Big((c_a, l_a), (c_b, l_b) \Big) \succ_{(\{a,b\})} \Big((c'_a, l'_a), (c'_b, l'_b) \Big)$$

and this contradicts **Responsibility**, which completes the proof for lemma 2. ■

Proof of Theorem 1. The proof of theorem 1 is immediate from the combination of Lemma 1 and Lemma 2 with the characterization of the maximin ordering by Hammond (1976).

□

B Results for the case $\underline{w} > 0$

B.1 Theoretical results

All the results in the main body of the paper have been presented with a worst wage rate in the economy equal to zero $\underline{w} = 0$. In this section, I show that the results can be extended to the case where $\underline{w} > 0$. This subsection derives the equivalent results as those presented in section 4 in the main text. It will not be assumed here that *Hardworking Poor Existence* holds. Rather, I will impose *Minimality* which restricts the number of tax-benefit systems (τ, D) that decentralizes a particular incentive-compatible allocation (c, l) by focusing on those where no inconsequential tax cut are left. It formally requires that the second-best budget $B(\tau, D, \underline{w}, \underline{h})$ coincides with the envelope curve of agents' indifference curves at (c, l). It is enough to restrict this assumption only for agents with the worst endowments.

Assumption (Minimality). *For $e \in E$ where there is at least one agent j with $(w_j, h_j) = (\underline{w}, \underline{h})$, a tax-benefit system (τ, D) that decentralizes the incentive-compatible (c, l) satisfies Minimality if*

$$B(\tau, D, \underline{w}, \underline{h}) = cl\left\{ \bigcup_{i:(w_i,h_i):(\underline{w},\underline{h})} Q_i(c_i, l_i) \right\}$$

where cl denotes the closure of a set, $Q_i(c_i, l_i)$ is the upper contour set of agent i at bundle (c_i, l_i) i.e. $Q_i(c_i, l_i) = \{(c_i', l_i') | (c_i', l_i') \succsim_i (c_i, l_i)\}$.

When the tax-benefit system is not minimal, one can devise tax cuts that do not affect any individual nor the budget constraint of the government. It is therefore a quite natural assumption. Figure A.4 provides an example of a violation of Minimality. Although (τ, D) decentralizes (c, l) in this two-agent economy, one could find a tax cut such that no one is affected. It would amount to make the blue locus coincide with the envelope of agents' indifference curves, i.e. to annihilate the shaded area in Figure A.4.

It is worth noting that when (τ, D) satisfies *Minimality*, $y - \tau(y)$ is non-decreasing in y because preferences are monotonic. As a consequence, *Minimality* forbids $\frac{\partial \tau(y)}{\partial y} > 1$ on some $y \in [0, \underline{w}]$, i.e. a confiscatory tax rate on some interval of low incomes. The next theorem is reminiscent to Theorem 2 in the main text.

Theorem A.2. *Under Minimality, for $e \in E$ such that $\underline{w} > 0$, and any two incentive-compatible allocations (c, l) and (c', l') decentralized by minimal (τ, D) and (τ', D') respectively, one has that (c, l) is socially preferred to (c', l') with respect to the FSOF whenever*

$$\min\left\{\underline{h} + D - \tilde{h}; \min_{0 \le y \le \underline{w}} (1 - \frac{\tilde{w}}{\underline{w}})y - \tau(y)\right\} > \min\left\{\underline{h} + D' - \tilde{h}; \min_{0 \le y \le \underline{w}} (1 - \frac{\tilde{w}}{\underline{w}})y - \tau'(y)\right\} \tag{2}$$

Proof. By Minimality, we know that there will be at least an agent with endowments $(\succsim_j, \underline{w}, \underline{h})$ such that she is indifferent between a bundle $(c_j, l_j) \in B(\tau, D, \underline{w}, \underline{h})$ and

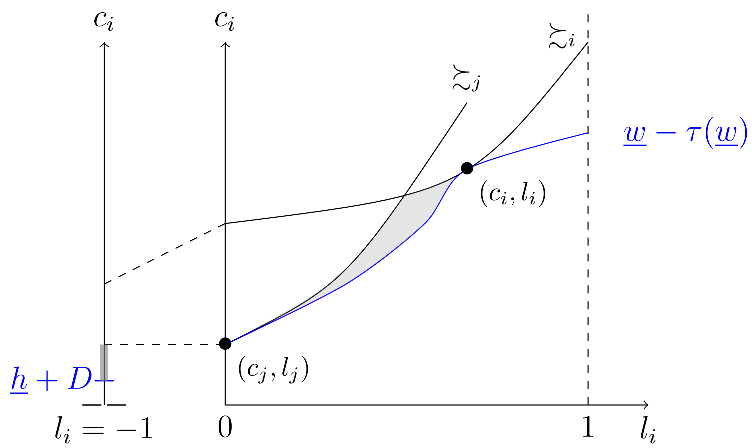

Figure A.4: Illustration of a tax-benefit system (τ, D) decentralizing (c, l) and violating Minimality. The shaded area corresponds to all points where the government could cut taxes at no cost: a tax cut (or benefit increase) in this region would expands the blue budget set $B(\tau, D\underline{w}, \underline{h})$ but neither j nor i would change their labor supply. Formally, the shaded area is the gap between the blue budget set and the closure of the upper contour set of agent's indifference curves at (c, l).

$\max_{\succsim_i} B(t, \tilde{w}, \tilde{h})$ for some t. One computes her welfare $M_j(c_j, l_j)$ as the smallest vertical translation t of the reference budget $B(0, \tilde{w}, \tilde{h})$ so that it coincides with one point of $B(\tau, D, \underline{w}, \underline{h})$, that is

$$\min_{j:(w_j,h_j)=(\underline{w},\underline{h})} M_j(c_j, l_j) = \min_{\hat{l}} t \text{ s.t. } \begin{cases} \underline{w}\hat{l} - \tau(\underline{w}\hat{l}) = \tilde{w}\hat{l} + t & \text{if } \hat{l} \in [0, 1], \\ \underline{h} + D = \tilde{h} + t & \text{if } \hat{l} = -1 \end{cases}$$

This is equivalent to

$$\min_{j:(w_j,h_j)=(\underline{w},\underline{h})} M_j(c_j, l_j) = \min \left\{ \underline{h} + D - \tilde{h}; \min_{0 \le y \le \underline{w}} (1 - \frac{\tilde{w}}{\underline{w}})y - \tau(y) \right\}$$

The remainder of the proof follows the same arguments as in the main text, *mutatis mutandis*. ■

The main difference with the main-text Theorem 2 is that now the worst-off in the labor market may be different from those earning the social assistance. In particular, they could be working with low pay, such as full time minimum wage earners.

Let me note that Minimality is less extreme than Hardworking Poor Existence as it does not require the existence of pathological preferences, but only imposes a mapping between the envelope of indifference curves and the tax-benefit system. If one assumes that there exist an agent that maximizes at each y such that there is no gap in the income distribution, Minimality is satisfied. Fortunately, the main text results for the case $\underline{w} = 0$ can be derived with Minimality rather than Hardworking Poor Existence. It simply consists in taking the limit of the sufficient statistic of Theorem A.2. when $\underline{w} \to 0$.

We conclude this subsection by deriving the optimal inactivity benefit with a formula reminiscent to Theorem 3.

Theorem A.3. *Under Minimality, for $e \in E$ with $\underline{w} > 0$, if there exists an incentive-compatible allocation decentralized by (τ^*, D^*) that is optimal with respect to the FSOF, then*

$$D^* = \tilde{h} - \underline{h} + \min_{0 \leq y \leq \underline{w}} (1 - \frac{\tilde{w}}{\underline{w}})y - \tau^*(y)$$

To sketch the optimal tax system, consider again the Laissez-faire policies $(\tau, D) = (0, 0)$. The government computes the smallest well-being in each sector using Theorem A.2. If it sets marginal tax rates such that $\frac{\partial \tau(y)}{\partial y} = 1 - \frac{\tilde{w}}{\underline{w}}$ for all $y \in (0, \underline{w})$, i.e. negative marginal tax rates on low incomes, then all low-incomes have identical well-being. Then, the government collects taxes as much as efficiency permits and sets the relative level of D^* and $-\tau^*(0)$ as in Theorem A.3. Again, this is very close to the optimal tax schedule of Fleurbaey and Maniquet (2007), where the present paper adds the key difference between D^* and $-\tau^*(0)$.

B.2 Empirical results

This subsection extends the results of section 5 in the main text to the case where $\underline{w} > 0$. I have assumed that the worst-off active is a minimum wage earner working full time, i.e. $\underline{w} = \arg\min_{0 \leq y \leq \underline{w}} (1 - \frac{\tilde{w}}{\underline{w}})y - \tau^*(y)$ for all countries. Apart from this, the simulation is run for 2019 with the exact same parameters as in the main text. The results are reported in Table A.1.

I conclude that the results are qualitatively similar even if quantitatively different. The policy conclusion is unchanged: governments should first increase support for low-income workers before any dollar of inactivity benefit is welfare-improving.

C Welfare recipient stigma

In this section, I consider the case of a government that does not wish to hold agents fully responsible for their preferences because their disutilities of participation have been partly driven by the *welfare recipient stigma*. It captures the idea that some agents remain inactive and do not take up conditional social benefits because enduring the screening device of the government entails a mental burden, rooted in the stigma that societies attach to benefits recipients.

Consider the following structure for the disutility of participation d_i:

$$d_i(c_i) \equiv u_i(c_i, -1) - u_i(c_i, 0) = s_i + \delta_i(c_i)$$

Table A.1: Summary of results. Sufficient increase in $\underline{w} - \tau(\underline{w})$ for $D = 1$ to be a welfare-improving reform, in percentage of current $\underline{w} - \tau(\underline{w})$.

Country	Lone parents	Singles
Australia	102.75%	116.50%
Belgium	50.47%	112.84%
Bulgaria	74.41%	139.96%
Canada	118.73%	165.95%
Czech Republic	96.99%	147.41%
Estonia	53.97%	114.82%
France	46.91%	91.72%
Greece	158.22%	158.22%
Germany	132.58%	214.20%
Croatia	129.42%	129.42%
Hungary	137.39%	233.21%
Israel	52.77%	119.34%
Ireland	46.10%	111.43%
Japan	54.23%	179.36%
Lithuania	88.95%	157.29%
Latvia	43.39%	139.87%
Luxembourg	50.49%	110.96%
Malta	93.07%	138.38%
Netherlands	116.16%	142.99%
New Zealand	44.68%	59.68%
Poland	125.05%	165.15%
Portugal	107.27%	107.27%
Romania	185.09%	192.31%
Slovenia	11.61%	97.30%
Slovak Republic	77.42%	118.30%
Spain	89.53%	102.38%
Turkey	119.83%	126.20%
United Kingdom	83.42%	134.01%
United States	103.51%	289.04%

where $\delta_i \in [-s_i, +\infty)$ is the idiosyncratic taste parameter[49] and $s_i \geq 0$ is the value of stigma. . Observe that s_i has a money-metric interpretation: it is the maximal amount of consumption that an agent is willing to forgo in order to escape enduring the screening device of the government.

For clarity of the exposition, let me assume that there are only two draws of this stigma utility cost: $s_i \in \{S, 0\}$ with $S > 0$. Hence, there are only two different exposures to the stigma cost of conditionality in the population: those that do suffer from it and those that do not[50]. The social planner wishes to compensate for s_i as well as w_i and h_i, while holding responsible for δ_i and their willingness to work.

Rather than deriving rigorously the full axiomatization that this compensation for s_i would entail, I sketch the result by an example. Consider the case of two fraternal twin sisters, agents k and j. They are identical in every respect but they differ in their exposure to the stigma utility cost: agent k suffers from it such that $s_k = S$ while agent j do not and $s_j = 0$.

Consider the bundle (c_j, l_j) and (c'_j, l'_j) that are such that agent j is indifferent between the two, and they are consumed when inactive and unemployed, respectively. I sketch this setup in Figure A.5.

In red are drawn the indifference curves of agent k. Obviously, when she is unemployed and consumes (c'_j, l'_j) she suffers the stigma $s_k = S > 0$ associated to this labor market status. If one computes the $M_i(c_i, l_i)$ utility of these two agents when they consume (c'_j, l'_j), one gets that $M_k(c'_j, l'_j) = M_j(c'_j, l'_j) - S$. In other words, *when unemployed*, the well-being measure derived in previous sections already accounts for the fact that the agent suffering from a larger disutility of participation (here, coming from the stigma) have a lower well-being.

Now, the difference is when the agents are inactive and consume (c_j, l_j). In this case, the indifference curve of k lies above the one of j, and they only coincide at (c_j, l_j). If one applies the $M_i(c_i, l_i)$ utility to this situation, one gets that $M_k(c_j, l_j) = M_j(c_j, l_j)$, i.e. both agents have the same level of well-being, even if agent k *would* have suffered from the stigma *had she joined the labor market*.

A government that compensates for S should treat the well-being of k when inactive as if she had actually experienced this stigma cost. In other words, the well-being measure when inactive must be reduced by s_i with respect to the $M_i(c_i, l_i)$ utility. Hence, the

[49]The fact that δ_i can now have negative values reflect the possibility for some agents to derive non-pecuniary benefits from labor market participation (such as friendliness from colleagues for example) which could partly offset the burden S puts on them.

[50]If this partitioning in two sets is degenerate, we are back to the analysis of the previous sections as there is no inequality to compensate for. In other words, if all agents experience the same stigma cost of conditionality, or if none of them does, the main formulas are unaffected.

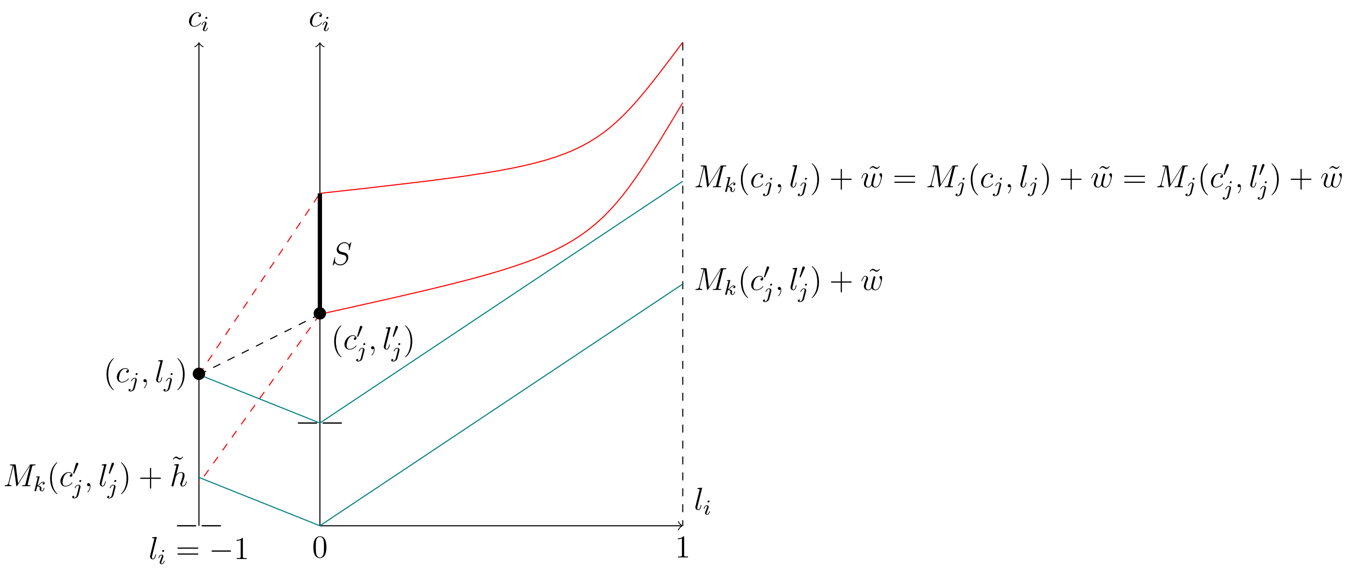

Figure A.5: Agents k and j are identical in every primitive but one: the inactive k suffers a stigma utility cost of S while the unemployed j does not. In red, the indifference curves of k. Agent j is indifferent between (c_j, l_j) and (c'_j, l'_j).

new well-being measure compensating for the stigma, is given by

$$W_i(c_j, l_j) = M_i(c_i, l_i) - (1 - a_i)s_i$$

In turn, all the analysis above can be repeated using $W_i(c_j, l_j)$ instead of $M_i(c_i, l_i)$. Obviously, the worst-off in the home sector will now have a well-being measure of $\underline{h} - S + D - \tilde{h}$. Hence, the optimal (τ^{**}, D^{**}) that maximins $W_i(c_j, l_j)$ follows:

$$D^{**} = S + \tilde{h} - \underline{h} + \min_{0 \leq y \leq \underline{w}} (1 - \frac{\tilde{w}}{\underline{w}})y - \tau^{**}(y)$$

I conclude that S positively influences the optimal inactivity benefit in an additive fashion and thereby weakens the conflict between fairness and basic income outlined in previous sections. Because of the formalization, S has a money-metric interpretation and should measured as an answer to this question: *how much would one be willing to pay (i.e. forgo consumption) to escape enduring the screening device of the government?*

To the best of my knowledge, such empirical estimates for S are not available in the literature. In order to justify that one dollar of inactivity benefit is welfare-improving, S must be at least larger than the estimates of table 1. In many instances, it seems unrealistically large. Alternatively, S could also be interpreted as an ethical parameter. In that case, estimates from table 1 would provide lower bounds on the government's willingness to pay to escape its own screening device.

D Details on the empirical application

I used the OECD tax-benefit simulator version 2.5.0. with the following parameters:

- Childless single 2020: aged 40, unemployed for 6 months, with 216 months of social security contributions accumulated over the lifetime, earning social assistance, for the year 2020. Eligible to social assistance, net of income tax and social security contributions.

- Lone parents 2020: aged 40, unemployed for 6 months, with 216 months of social security contributions accumulated over the lifetime, earning social assistance, for the year 2020. Children aged 4 and 6. Eligible to social assistance, lone parents support, net of income tax and social security contributions.

- Childless single 2019: aged 40, unemployed for 6 months, with 216 months of social security contributions accumulated over the lifetime, earning statutory minimum wage. Eligible to social assistance, in-work benefits, net of income tax and social security contributions.

- Lone parents 2019: aged 40, unemployed for 6 months, with 216 months of social security contributions accumulated over the lifetime, earning statutory minimum wage. Children aged 4 and 6. Eligible to social assistance, in-work benefits, lone parents support, net of income tax and social security contributions

The set of countries covered by the G-SWA survey (Aksoy et al., 2023) is a strict subset of the 29 countries I study. Whenever the estimate for F was not available, I kept the maximum of the series, i.e. 100 minutes per day.

The legal length of the working week is taken from the OECD (2021) (Annex Table 5.A.1.). The statutory length may differ from the negotiated length in some countries. When both are present, I took the maximum among the two. If both are absent, I set the length of the working week to 45 hours, corresponding to the maximum of the series. I considered that the working week lasts 5 days.

Appendix references

Aksoy, C. G., Barrero, J. M., Bloom, N., Davis, S. J., Dolls, M., & Zarate, P. (2023). Time savings when working from home. *National Bureau of Economic Research Working Papers*.

Bosmans, K., Decancq, K., & Ooghe, E. (2018). Who's afraid of aggregating money metrics? *Theoretical Economics*, *13*(2), 467–484.

Fleurbaey, M., & Maniquet, F. (2006). Fair income tax. *The Review of Economic Studies*, *73*(1), 55–83.

Fleurbaey, M., & Maniquet, F. (2007). Help the low skilled or let the hardworking thrive? a study of fairness in optimal income taxation. *Journal of Public Economic Theory*, *9*(3), 467–500.

Fleurbaey, M., & Maniquet, F. (2011). *A theory of fairness and social welfare*. Econometric Society Monograph (vol 48). Cambridge University Press.

Fleurbaey, M., & Maniquet, F. (2018). Optimal income taxation theory and principles of fairness. *Journal of Economic Literature*, *56*(3), 1029–79.

Hammond, P. J. (1976). Equity, arrow's conditions, and rawls' difference principle. *Econometrica: Journal of the Econometric Society*, 793–804.

Hansson, B. (1973). The independence condition in the theory of social choice. *Theory and Decision*, *4*(1), 25–49.

OECD. (2021). *Oecd employment outlook*. OECD Publishing: Paris, France.

Piacquadio, P. G. (2017). A fairness justification of utilitarianism. *Econometrica*, *85*(4), 1261–1276.

Valletta, G. (2014). Health, fairness and taxation. *Social Choice and Welfare*, *43*(1), 101–140.